

Data Visualization:

A Practical Guide

Danielle Presgraves, Ph.D.

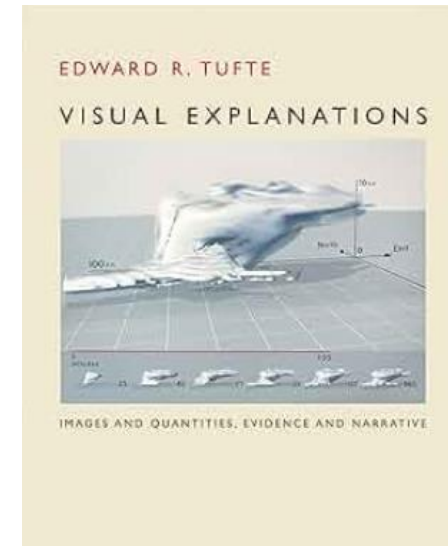
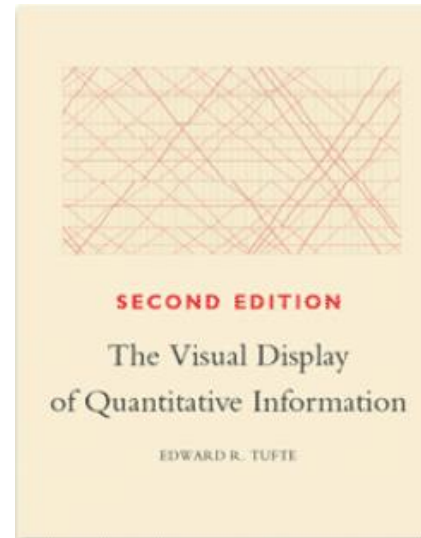
Learning Objectives:

1. Classify the variable type (categorical, ordinal, numeric, etc.)
2. Identify the appropriate visual representation based on variable and question
3. Integrate storytelling to emphasize data understanding, not just presentation
4. Prioritize accuracy, ethics, and honest communication
5. Simplify to reduce cognitive load on your audience

Agenda:

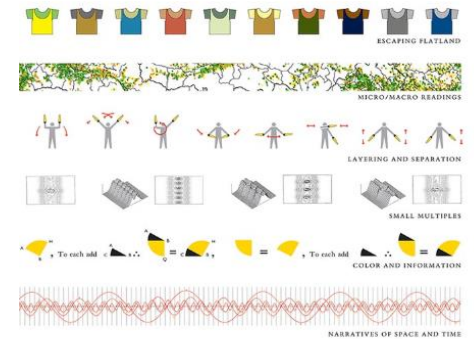
- 1. Visualization can fail and succeed.**
2. Find a narrative for your audience: The Hero's Journey
3. Choosing the correct visualization:
 - 1.(Brief) Historic visualization visionaries
 - 2.Technical Aspects
 - 3.Artistic & human factors choices

Edward Tufte

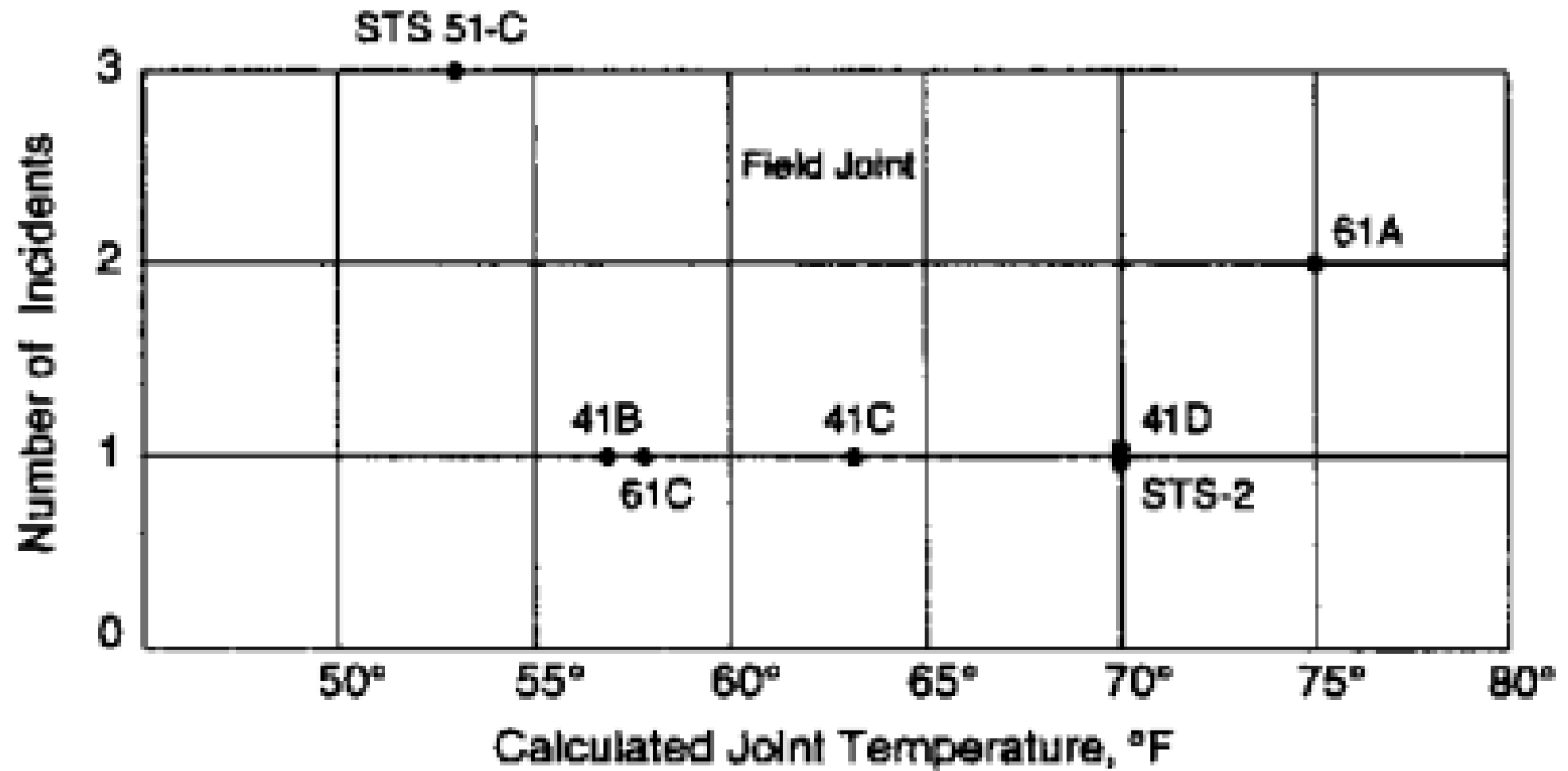


Edward R. Tufte

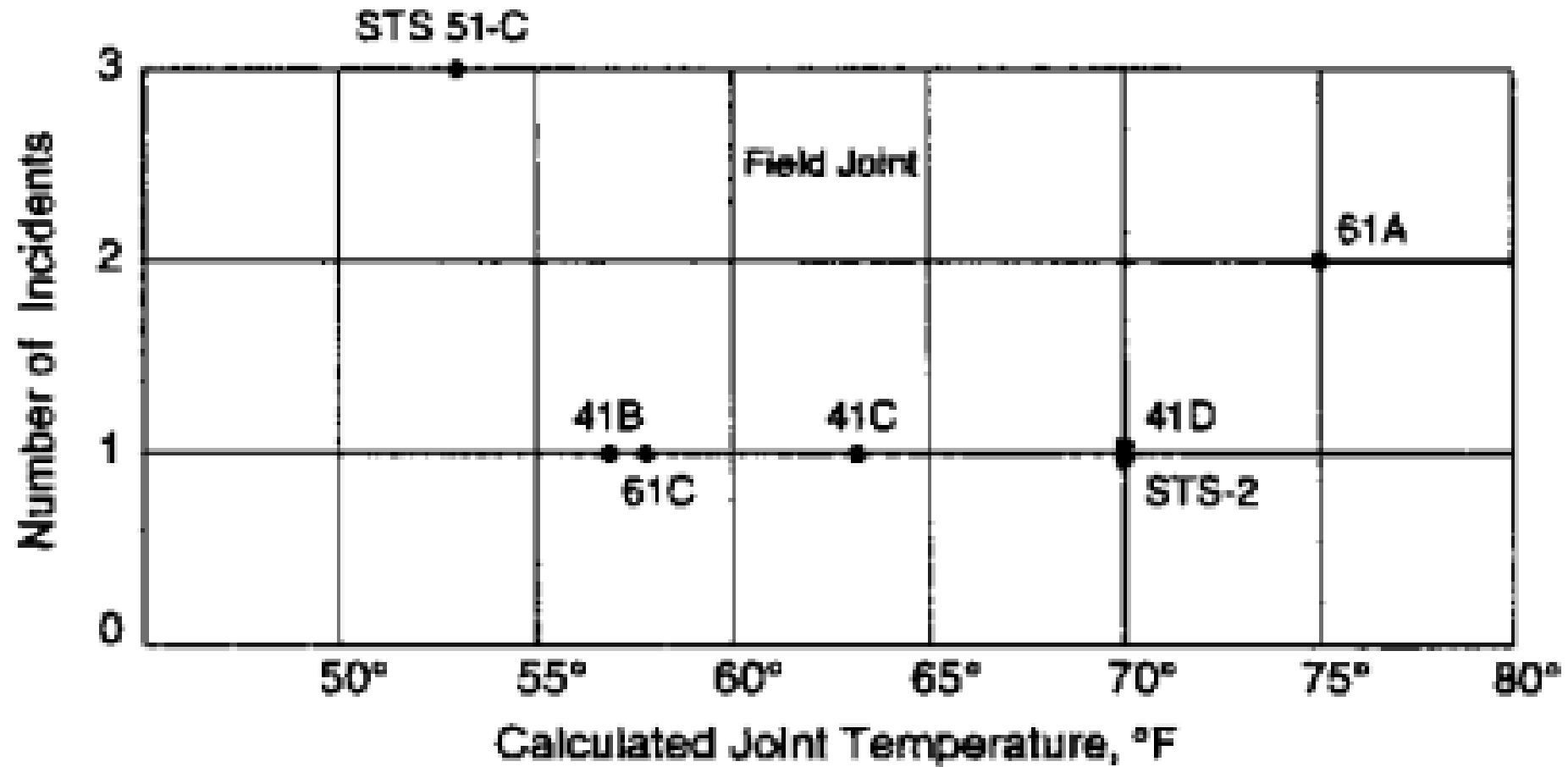
Envisioning Information



Why visualization matters



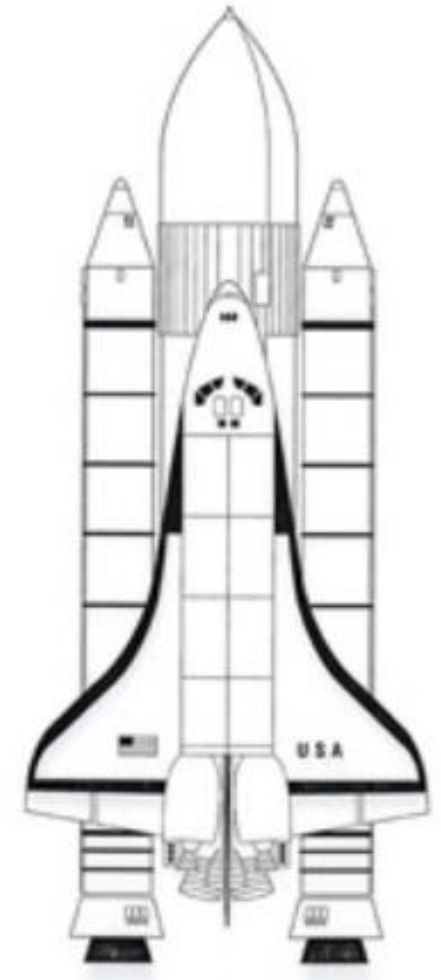
Post-launch diagram*



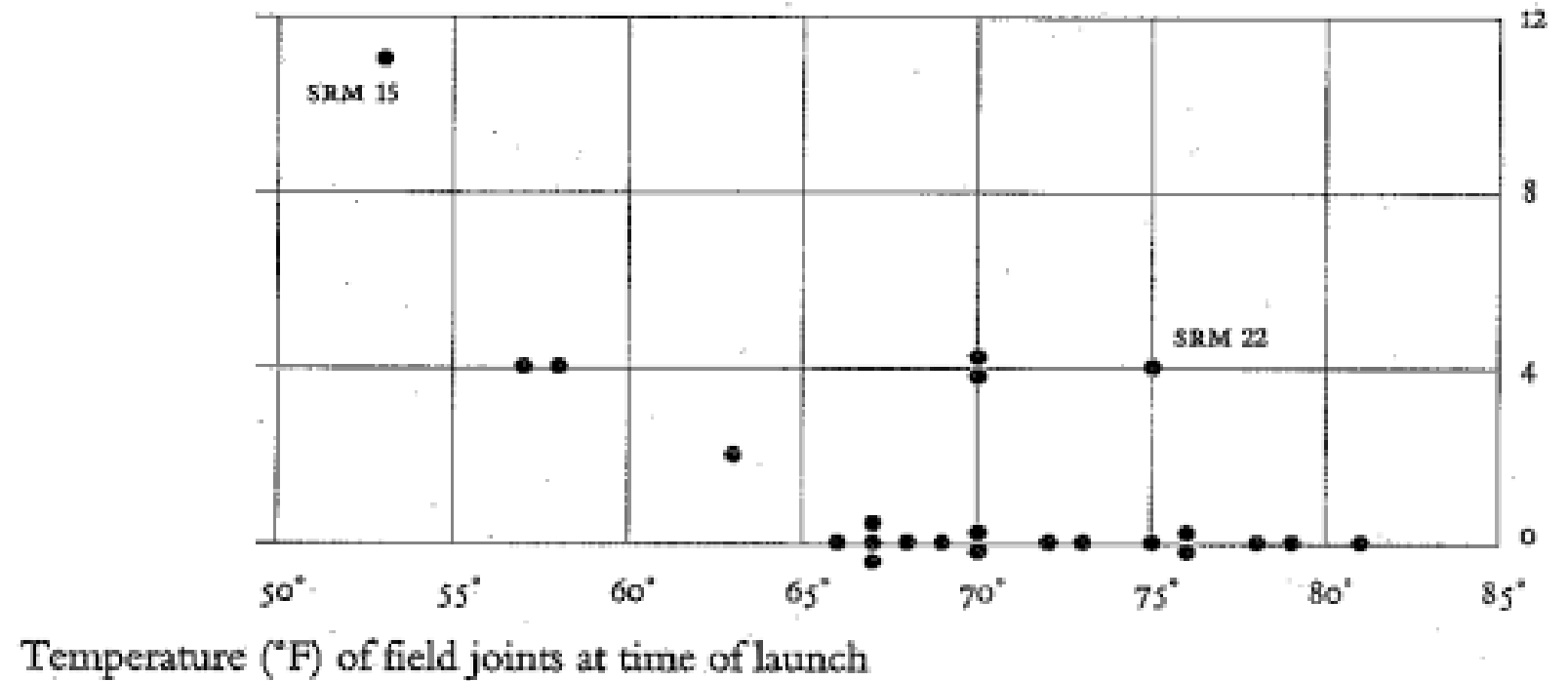
* drawn after incident by a lawyer and executive director

The Challenger pre-launch presentation from concerned engineers had 13 charts, including:

Flight	Date	Temperature °F	Erosion incidents	Blow-by incidents	Damage index	Comments
51-C	01.24.85	53°	3	2	11	Most erosion any flight; blow-by; back-up rings heated.
41-B	02.03.84	57°	1		4	Deep, extensive erosion.
61-C	01.12.86	58°	1		4	O-ring erosion on launch two weeks before Challenger.
41-C	04.06.84	63°	1		2	O-rings showed signs of heating, but no damage.
1	04.12.81	66°			0	Coollest (66°) launch without O-ring problems.
6	04.04.83	67°			0	
51-A	11.08.84	67°			0	
51-D	04.12.85	67°			0	
5	11.11.82	68°			0	
3	03.22.82	69°			0	
2	11.12.81	70°	1		4	Extent of erosion not fully known.
9	11.28.83	70°			0	
41-D	08.30.84	70°	1		4	
51-G	06.17.85	70°			0	
7	06.18.83	72°			0	
8	08.30.83	73°			0	
51-B	04.29.85	75°			0	
61-A	10.30.85	75°		2	4	No erosion. Soot found behind two primary O-rings.
51-I	08.27.85	76°			0	
61-B	11.26.85	76°			0	
41-G	10.05.84	78°			0	
51-J	10.03.85	79°			0	
4	06.27.82	80°			?	O-ring condition unknown; rocket casing lost at sea.
51-F	07.29.85	81°			0	



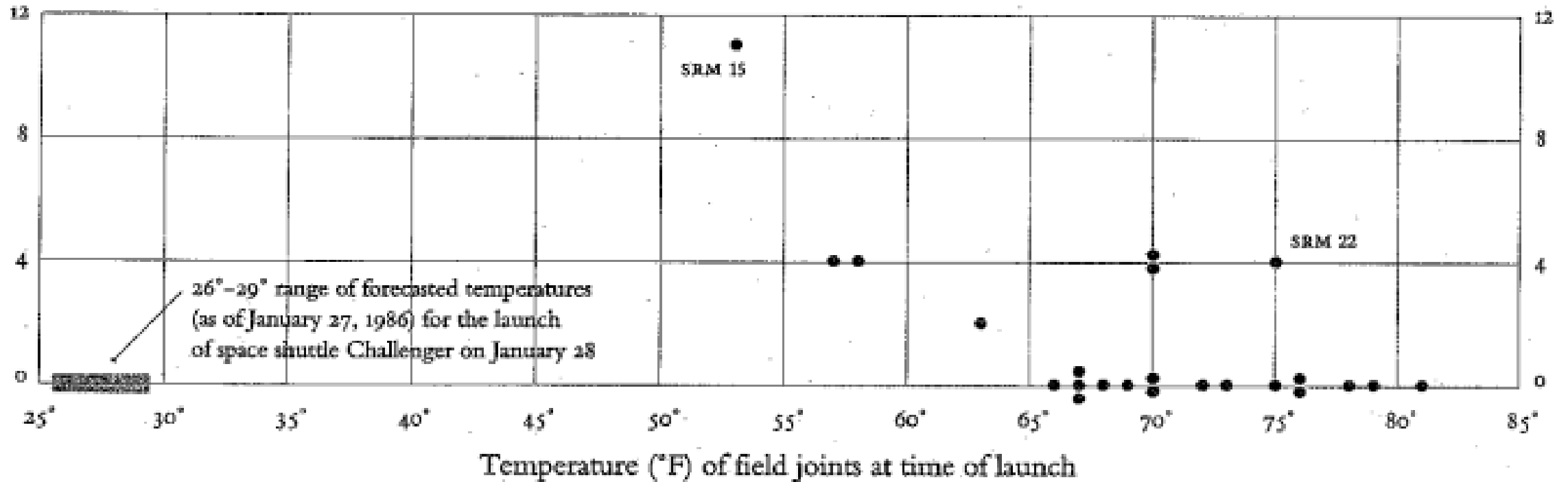
Post-launch Redrawn by Tufte



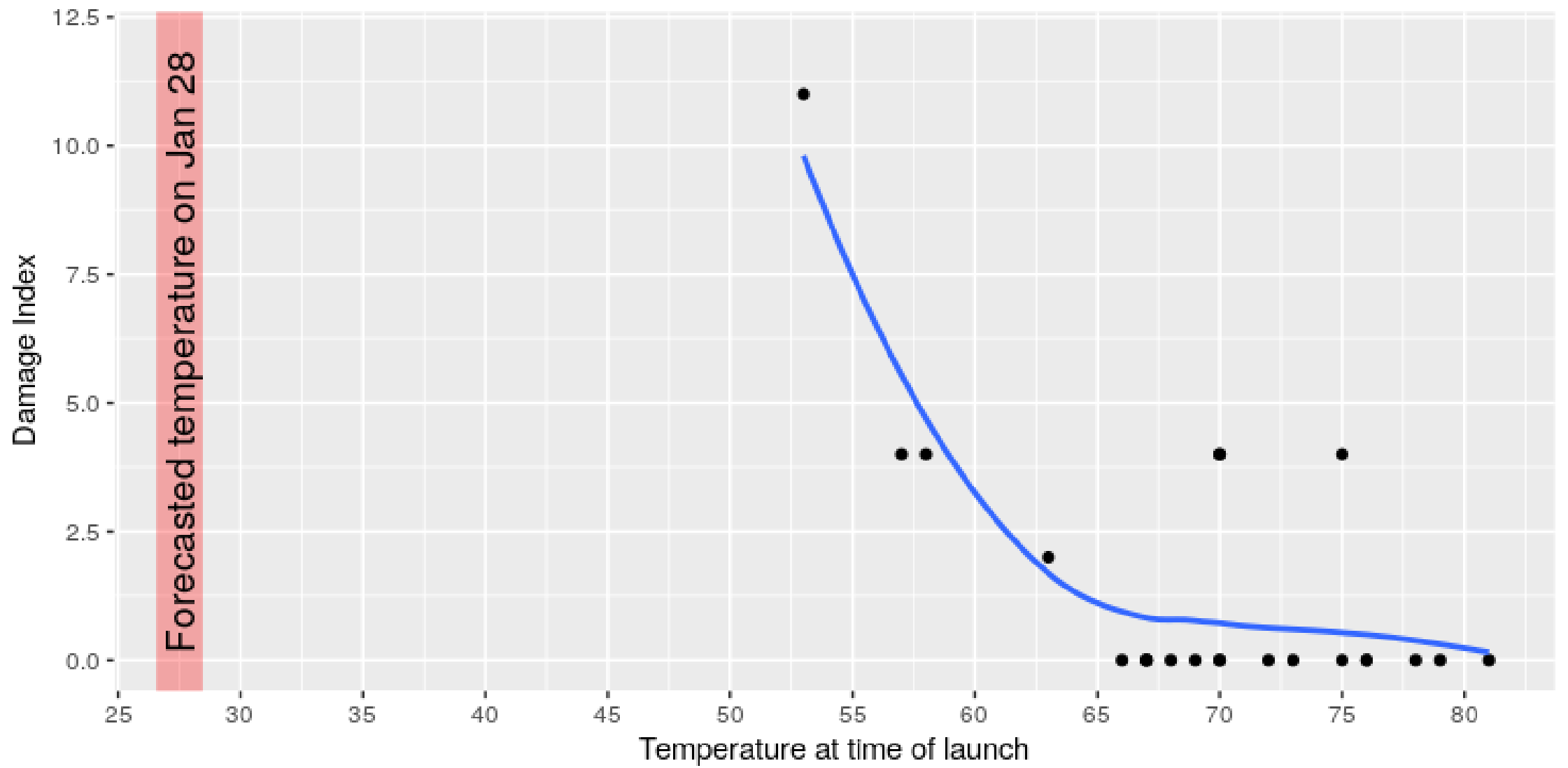
Tufte, Visual and Statistical Thinking (1997)

Post-launch Redrawn by Tufte

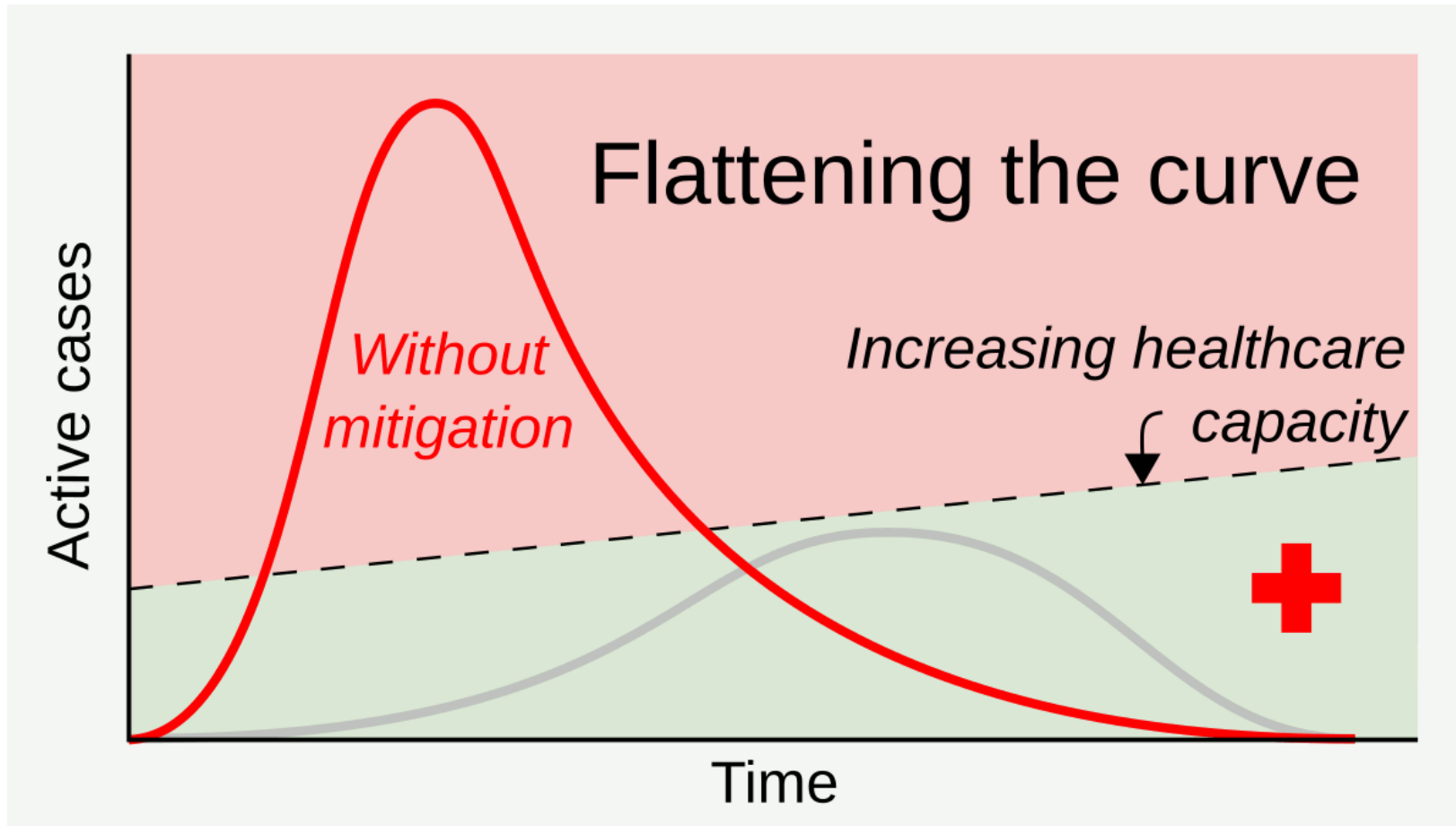
O-ring damage
index, each launch



Tufte, Visual and Statistical Thinking (1997)



Flatten the curve.

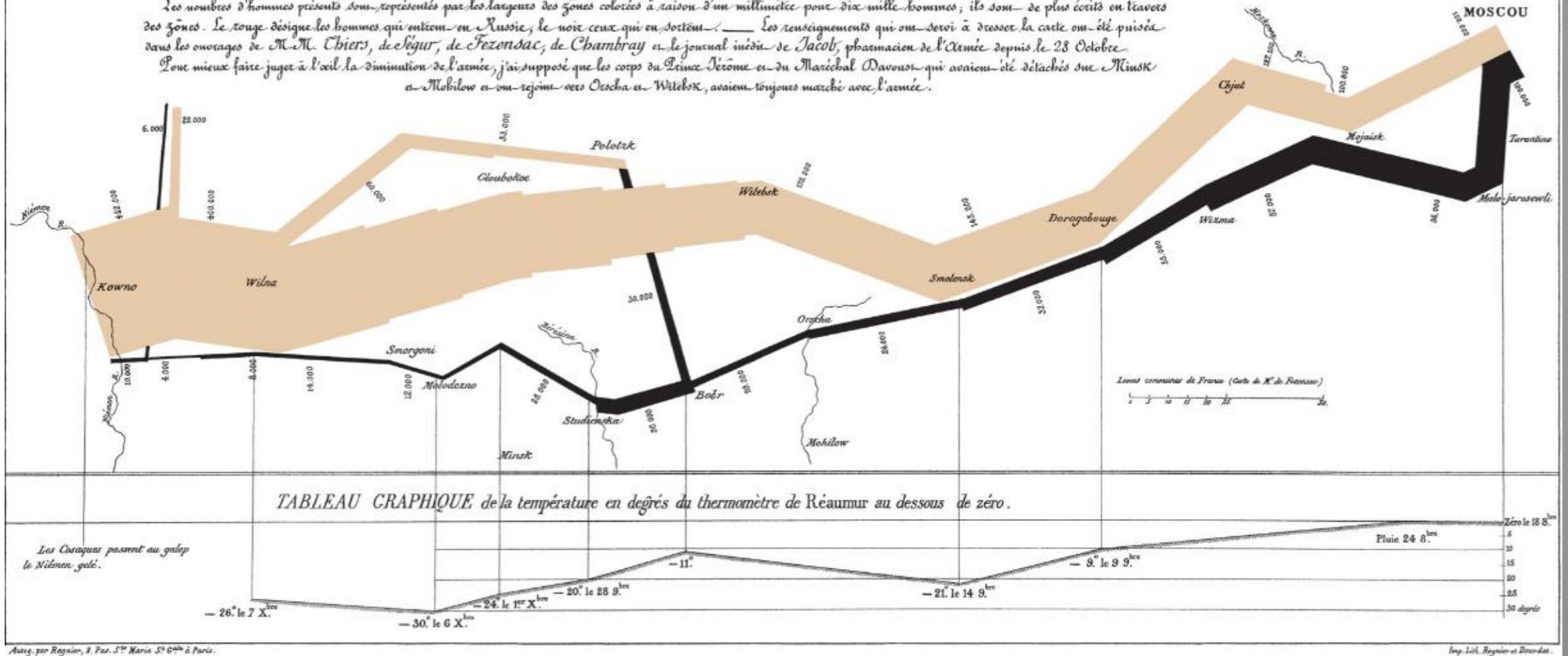


Good figures can change public behavior.

Carte Figurative des pertes successives en hommes de l'Armée Française dans la campagne de Russie 1812-1813.

Dressée par M. Minard, Inspecteur Général des Ponts et Chaussées en retraite Paris, le 20 Novembre 1869.

Les nombres d'hommes présents sont représentés par les largueurs des zones colorées à raison d'un millimètre pour dix mille hommes; ils sont de plus écrits en travers des zones. Le rouge désigne les hommes qui ont été en Russie; le noir ceux qui en sont sortis. Les renseignements qui ont servi à dresser la carte ont été puisés dans les ouvrages de M. M. Thiers, de Ségur, de Fezensac, de Chambray et le journal inédit de Jacob, pharmacien de l'Armée depuis le 28 Octobre. Pour mieux faire juger à l'œil la diminution de l'armée, j'ai supposé que les corps du Prince Néssmé et du Maréchal Davoust qui avaient été détachés sur Minsk et Mohilow et qui rejoindront Otscha et Witebsk, avaient toujours marché avec l'armée.



“probably the best statistical graphic ever drawn” (Tufte, Visual Display of Quantitative Information)

Agenda:

1. Visualization can fail and succeed.
 - 2. Find a narrative for your audience: The Hero's Journey**
 3. Choosing the correct visualization:
 - 1.(Brief) Historic visualization visionaries
 - 2.Technical Aspects
 3. Artistic & human factors choices
-

Data Visualization requires **clear thinking about a result**

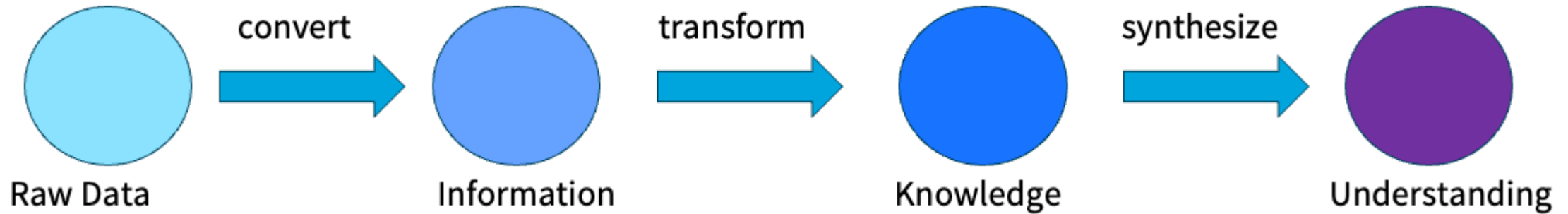
Narrative:

“A story is a set of observations, facts, or events, true or invented, that are presented in a specific order such that they create an emotional reaction in the audience. The emotional reaction is created through the build-up of tension at the beginning of the story followed by some type of resolution towards the end of the story.”

“**Most data visualization is done for the purpose of communication.** We have an insight about a dataset, and we have a potential audience, and we would like to convey our insight to our audience. To communicate our insight successfully, we will have to present the audience with a clear and exciting story. *The need for a story may seem disturbing to scientists and engineers, who may equate it with making things up, putting a spin on things, or overselling results*” - Claus Wilke

If you don't provide a clear story, the audience will make one up!

“The role of scientists is to collect data and **transform** them into understanding. **Their role as authors is to present that understanding.**”



“The data are supporting actors in the story you tell. The lead actors are the questions and the larger issues you are addressing. **The story grows from the data, but the data are not the story.**”

“Only by **exploring the boundaries** and limits of your **data** can you find the important story.”

THE HERO'S JOURNEY

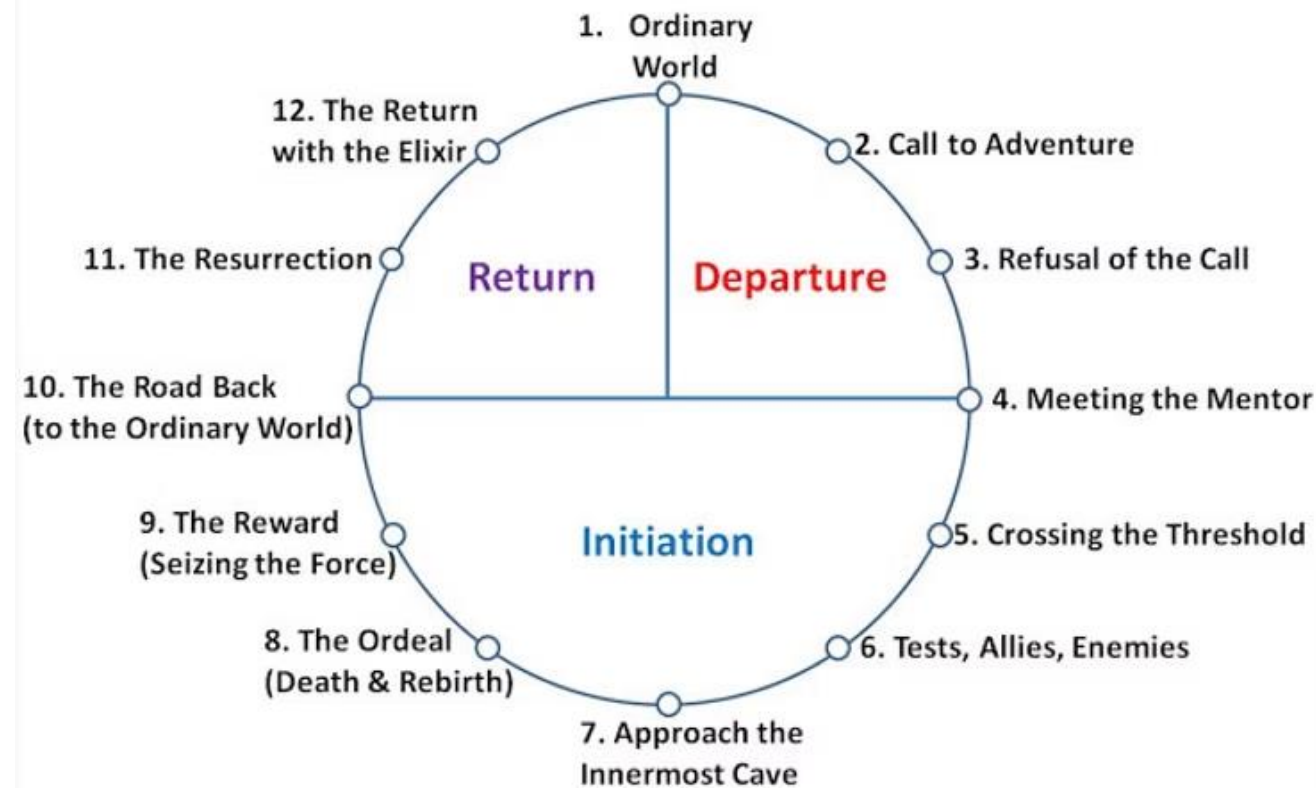


Image Source : mikeduran.com

1. Ordinary World: Set the baseline
2. Call to Adventure: An anomaly or pattern
3. Threshold: From data to implications
4. The Reward: Insight or discovery
5. The Return: Application or next step

You can also use a three-act structure

<https://www.storyboardthat.com/articles/e/heroic-journey>

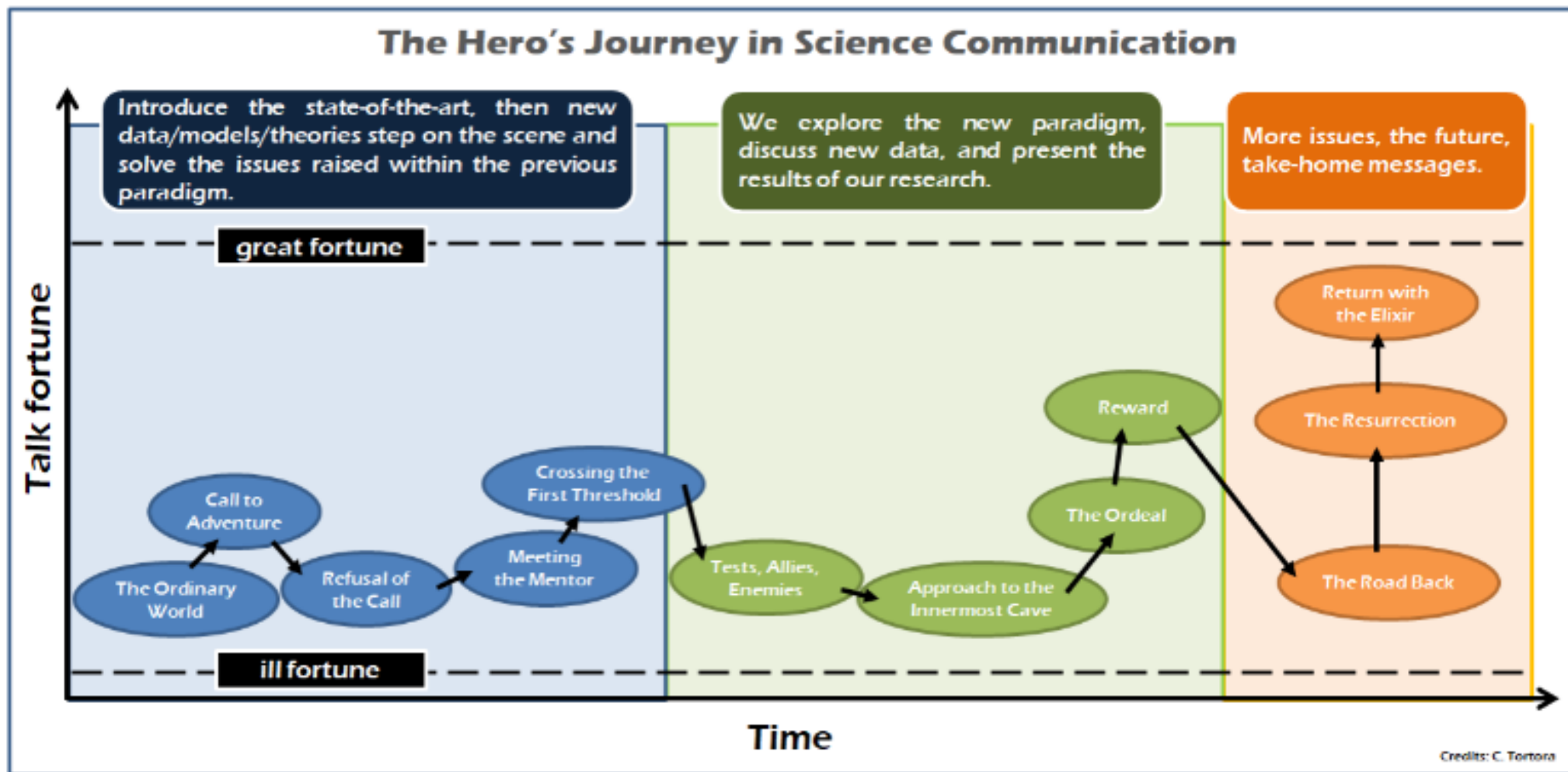


Figure 4. The Hero's Journey in Science communication in a nutshell. The talk emotional arc is presented on the Vonnegut's 'beginning-end' and 'ill fortune-great fortune' axes. The talk fortune is shown as a function of talk time, as a progression through lows and highs, divided in the 12 stages and the three acts: Departure or Separation (blue), Initiation (green) and Return (orange). Stages in the lowest part of the diagram correspond to issues/problems/failures, while on the top to discoveries and positive results. Note that the precise vertical position of each stage in the diagram is flexible. Credits: C. Tortora

Every research project tells a story:

1. What was known? [The Ordinary World](#)
2. What unexpected observation occurred? [Call to Adventure/Crossing the Threshold](#)
3. What questions did that create? [Trials and Revelations](#)
4. How was the mystery investigated? [Climax/Ordeal/Reward](#)
5. What was discovered? [Return with Elixir](#)
6. Why does it matter? [The World has changed](#)

[Question](#) ➡ [Investigation](#) ➡ [Discovery](#) ➡ [Impact](#)

Agenda:

1. Visualization can fail and succeed.
 2. Find a narrative for your audience: The Hero's Journey
 - 3. Choosing the correct visualization:**
 1. (Brief) Historic visualization visionaries
 2. Technical Aspects
 3. Artistic & human factors choices
-

Data Visualization Requires Clear Thinking

1. Technical Foundations

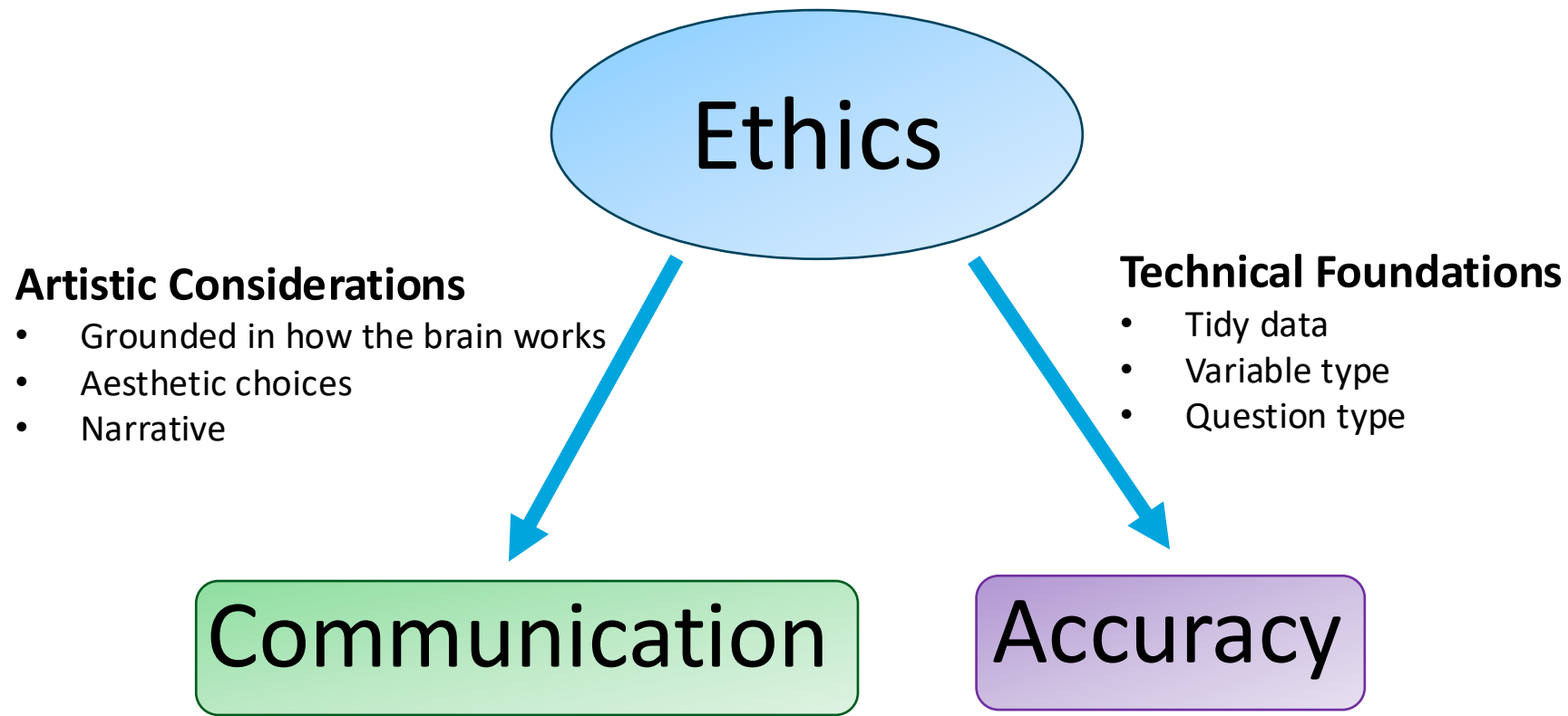
- Tidy data
- Variable type
- Question type



2. Artistic Considerations

- Grounded in how the brain works
- Aesthetic choices
- narrative

No coding in this presentation, but you might notice philosophical similarities to Leland Wilkinson “The grammar of visualization” and Hadley Wickham’s tidyverse/ggplot package



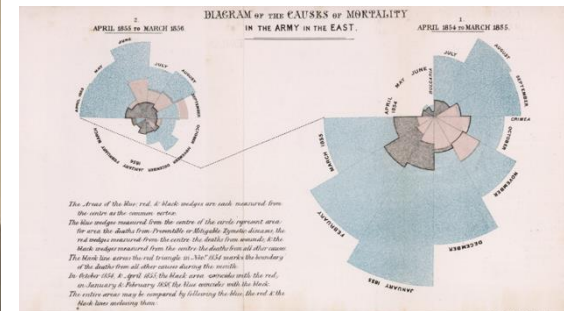
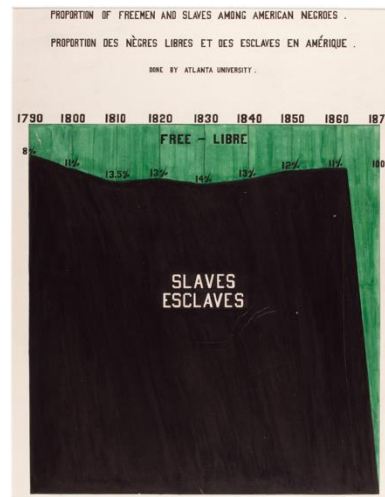
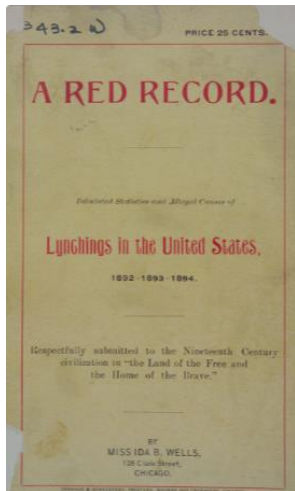
Tell an honest story

- Support conclusions with transparent methods
- Avoid distorted axes, cherry-picking, and hidden baselines

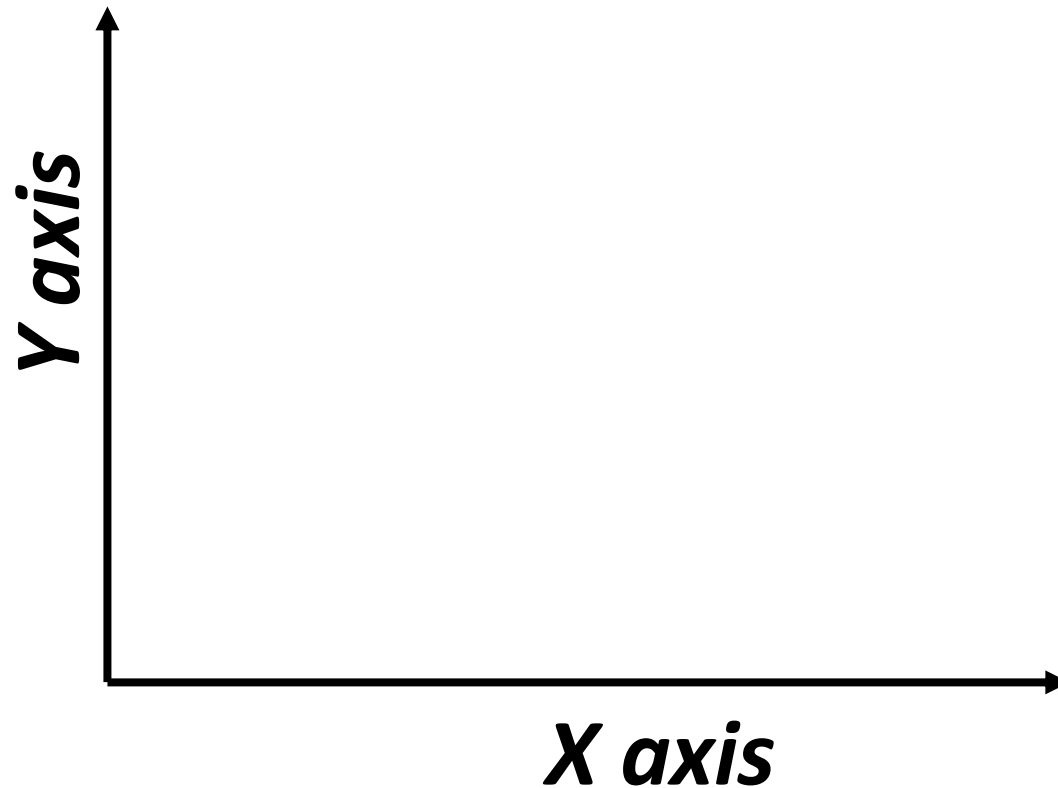
Ethics

Communication

Accuracy



Anatomy of a plot



- Axes: choose meaningful baselines; avoid dual axes unless essential
- Variable mapping: encode key data in position first

Anatomy of a plot

X axis = horizontal axis

Y axis = vertical axis

Variable type:

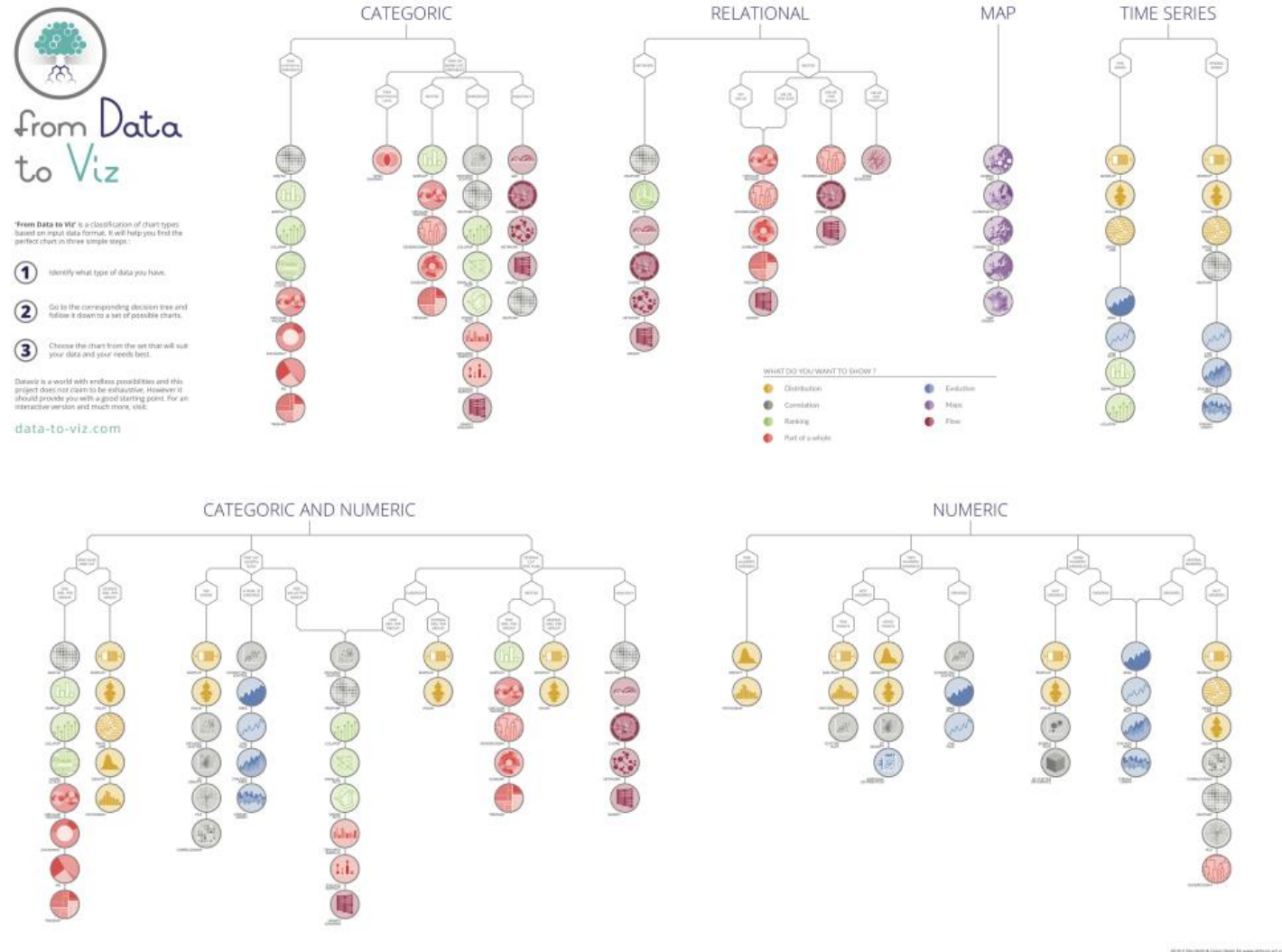
Categorical

- Ordinal: can be ordered
 ex. Age, Socio economics status, education level
- Nominal: cannot be meaningfully ordered
 ex. Religion, blood group, cause of death, treatments, dog breeds
- Binary
 ex. Success and Failure

Numeric

- Continuous: measure
 ex. Age, height, distance
- Discrete: count
 ex. Number of patients, results of rolling a die

Variable type governs plotting possibilities



Variable type (and number) governs plotting possibilities

# variables	Variable Type	Recommended Plots	Use Case
1 (univariate)	Categorical	Bar Plot, Pie Chart	Comparing category frequencies
	Numerical	Histogram, Density Plot	Understanding distribution
2 (Bivariate)	Categorical & Categorical	Grouped Bar Chart, Mosaic Plot	Comparing proportions of two groups
	Numerical & Categorical	Boxplot, Violin Plot, Strip Plot	Comparing distributions across categories
	Numerical & Numerical	Scatter Plot, Line Plot, Hexbin Plot	Examining relationships or trends
3+ (Multivariate)	Multiple Categorical	Stacked Bar Chart	Analyzing categorical interactions
	Multiple Numerical	Scatterplot Matrix	Comparing multiple numeric relationships
	Mixed	Faceted Plots, Heatmap, Bubble Chart	Visualizing mixed data relationships

Examples

A subset from a dataset of 200 mice:

Mouse_ID	Gene_Expression	Body_Weight	Age	Treatment_Group	Genotype	Survival_Status	Detailed_Genotype	Mouse_Strain
Mouse_1	57.45	26.79	8	Control	Knockout	Alive	AA	C57BL/6
Mouse_2	47.93	27.8	6	Drug_B	Knockout	Deceased	aa	BALB/c
Mouse_3	59.72	30.42	21	Drug_A	Wildtype	Alive	AA	BALB/c
Mouse_4	72.85	30.27	11	Drug_A	Wildtype	Alive	AA	C57BL/6
Mouse_5	46.49	18.11	22	Control	Knockout	Deceased	aa	BALB/c
Mouse_6	46.49	20.31	10	Drug_A	Knockout	Alive	AA	C57BL/6

How would we visualize the following?

1. Is there a relationship between **Gene_Expression** and **Body_Weight**?
2. How does **Gene_Expression** differ among **Treatment_Group**? Or Genotype? Or Mouse_Strain?
3. Does **Body_Weight** vary between different **Detailed_Genotype** groups? Or Mouse_Strain?
4. Is there an association between **Treatment_Group** and **Survival_Status**?

How would we visualize the following?

1. Is there a relationship between **Gene_Expression** and **Body_Weight**?



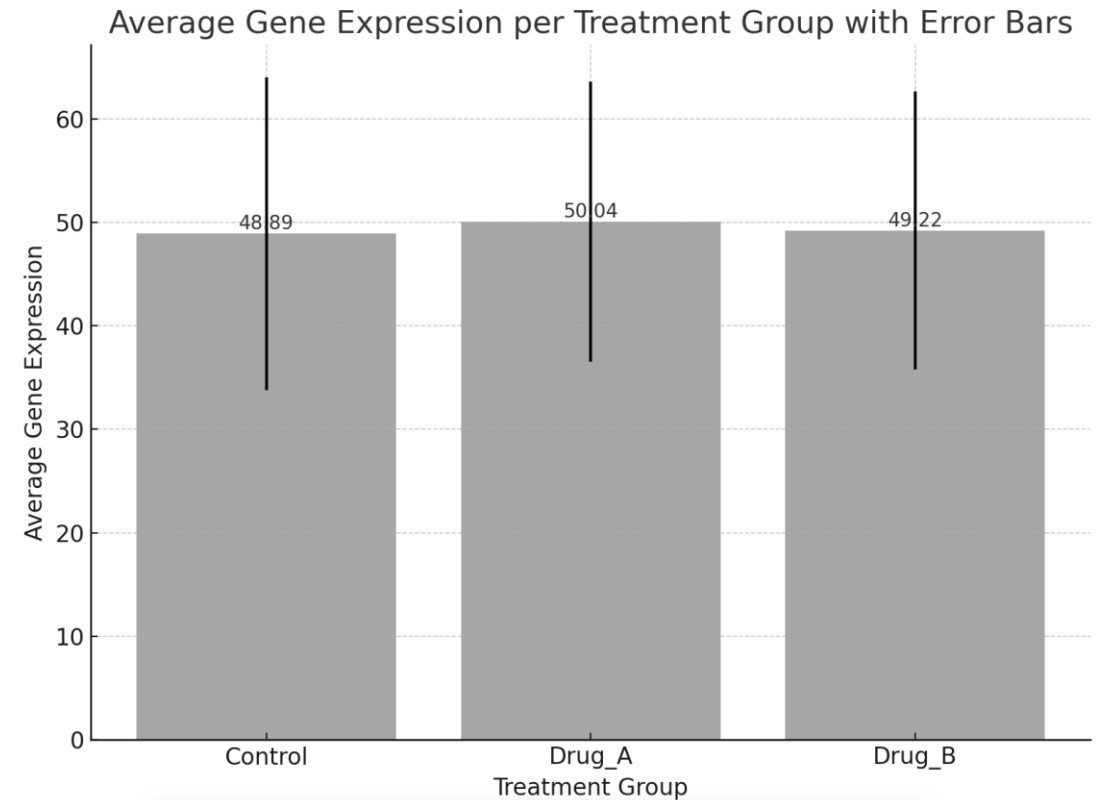
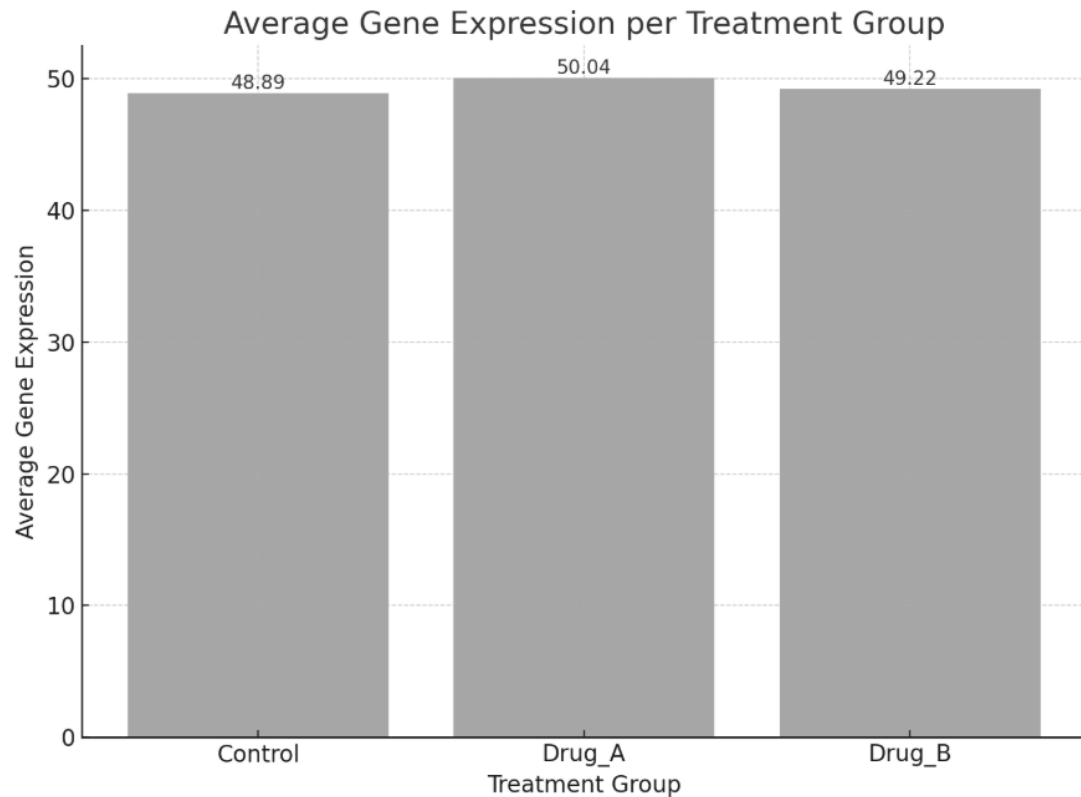
How would we visualize the following?

1. Is there a relationship between **Gene_Expression** and **Body_Weight**?



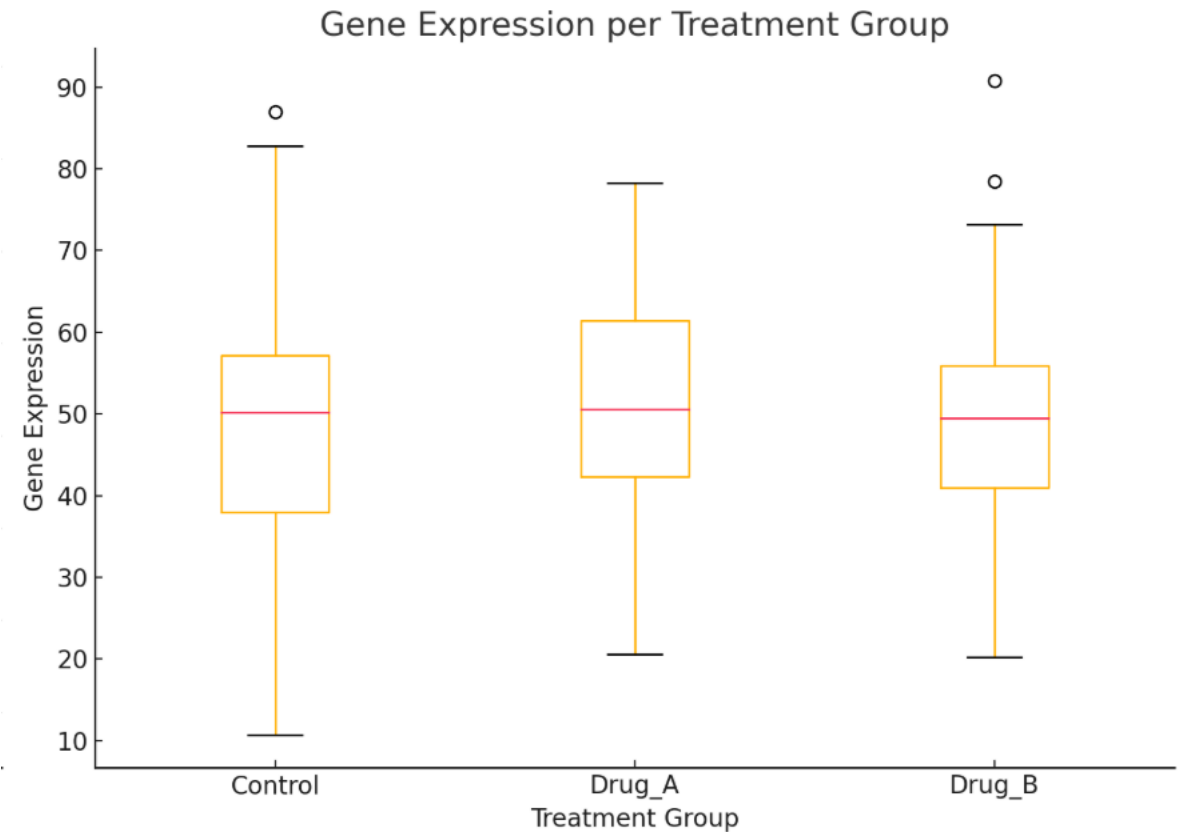
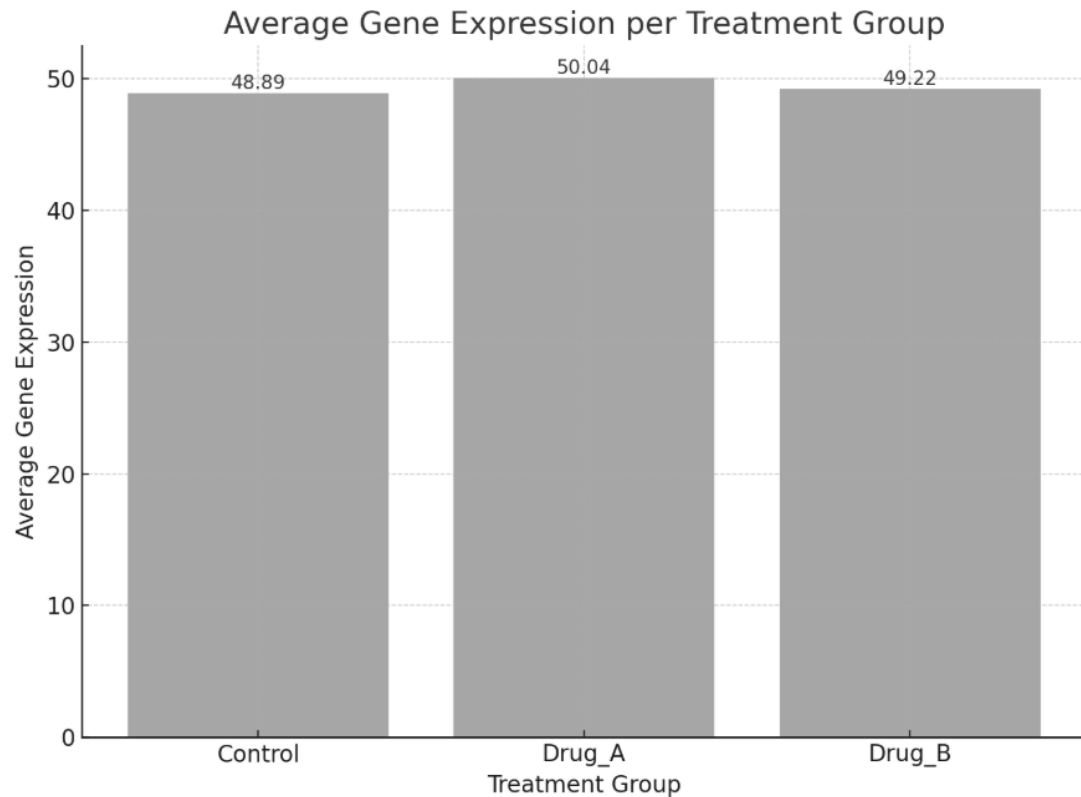
How would we visualize the following?

2. How does **Gene_Expression** differ among **Treatment_Group**? Or **Mouse_Strain**?



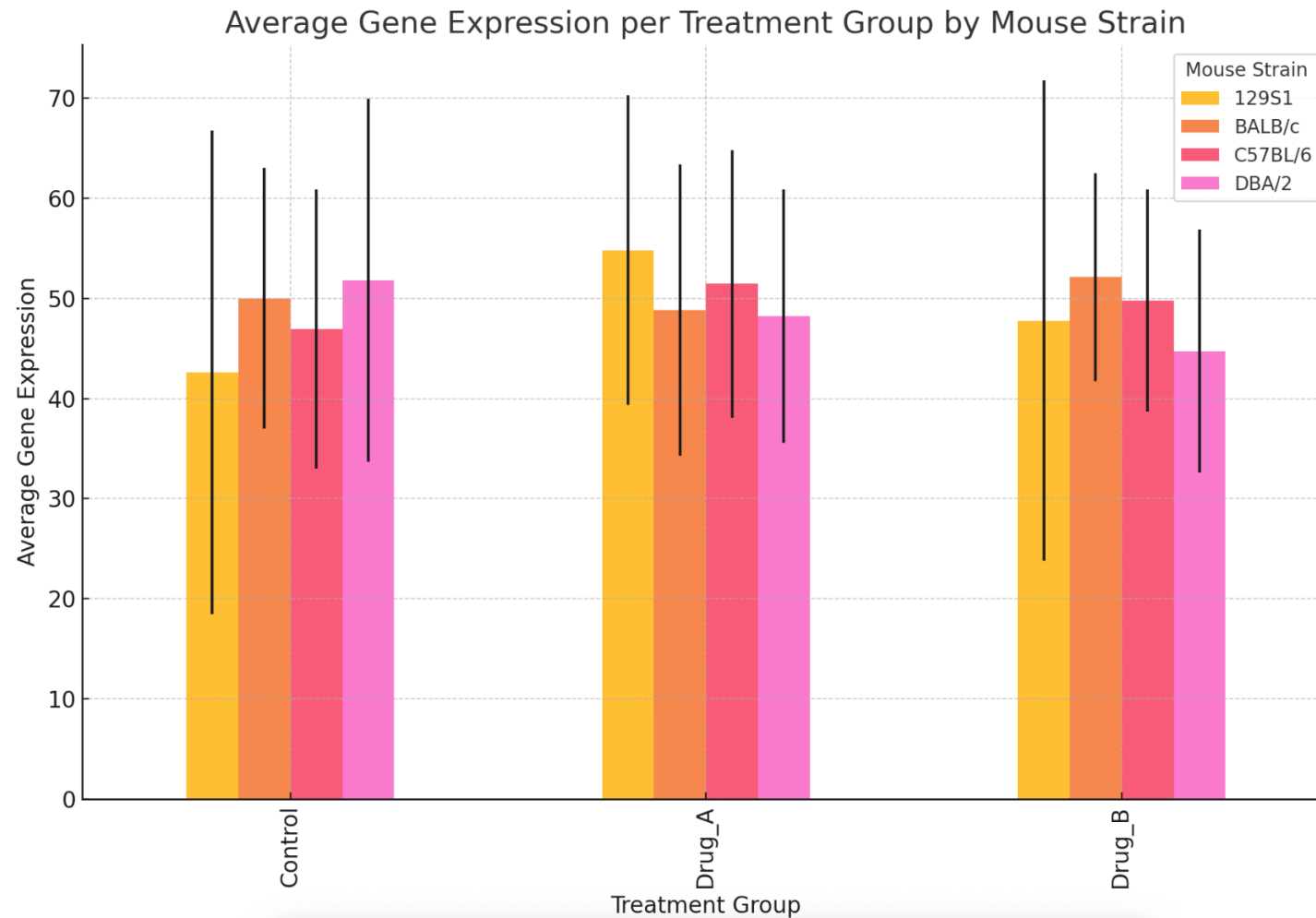
How would we visualize the following?

2. How does **Gene_Expression** differ among **Treatment_Group**? Or **Mouse_Strain**?



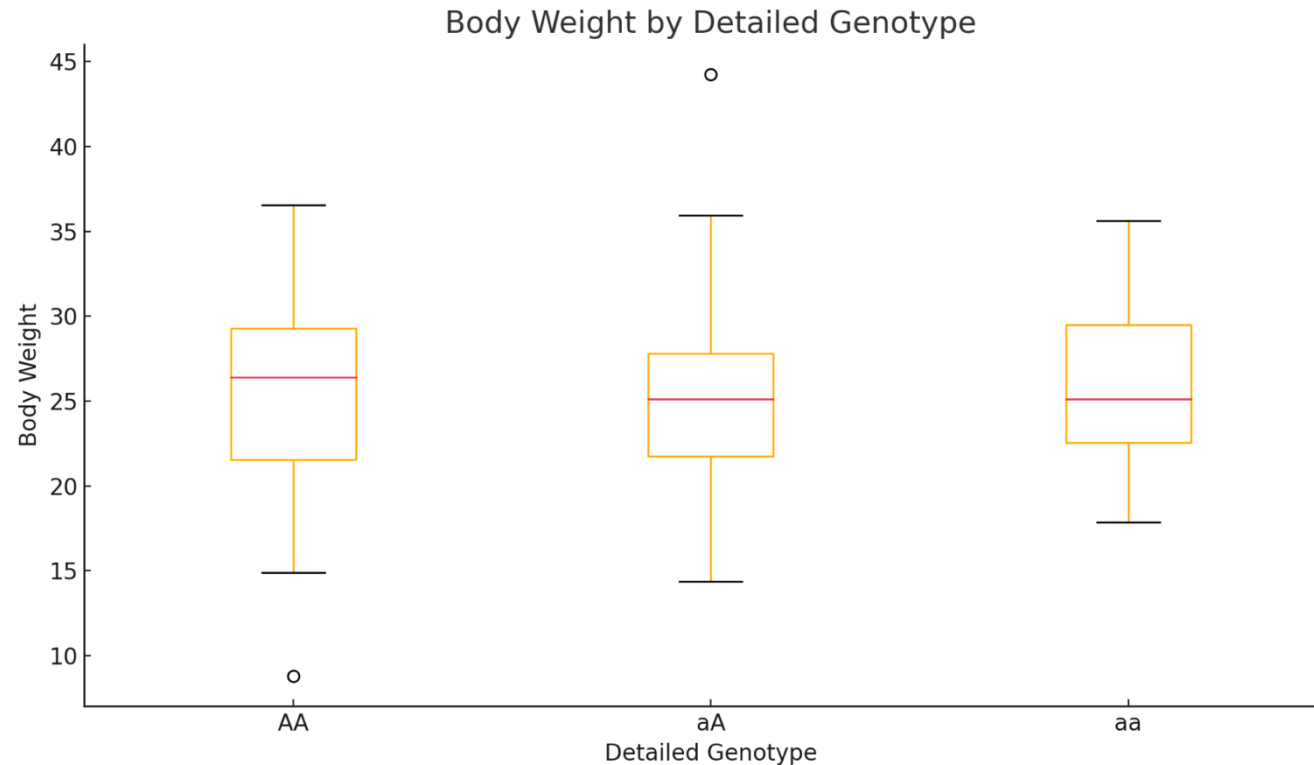
How would we visualize the following?

2. How does **Gene_Expression** differ among **Treatment_Group** and **Mouse_Strain**?



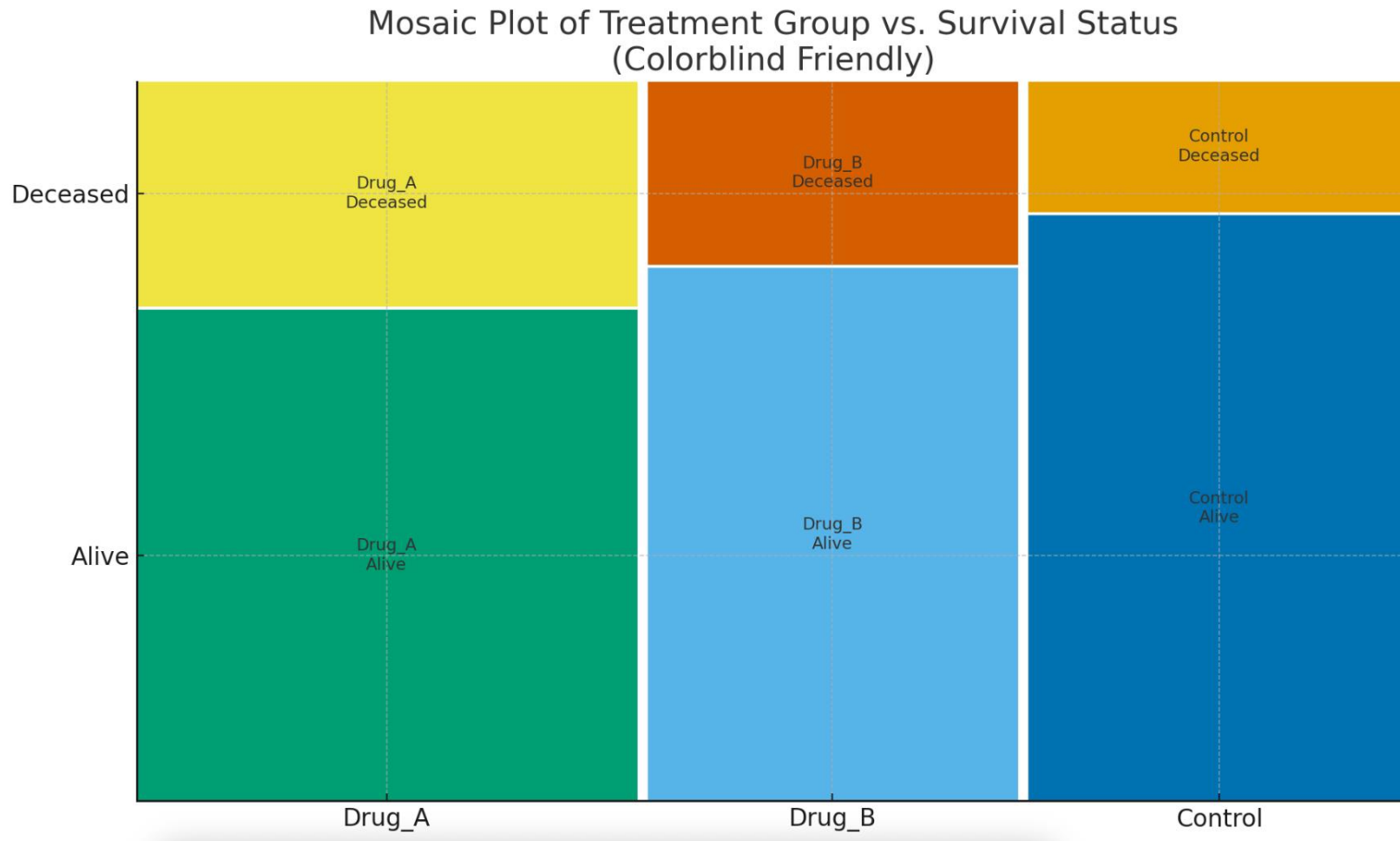
How would we visualize the following?

3. Does **Body_Weight** vary between different **Detailed_Genotype** group?



How would we visualize the following?

4. Is there an association between **Treatment_Group** and **Survival_Status**?



Once you have your variables and question, there
are still decisions to be made....

How Our Eyes Read Charts

Pre-attentive attributes (Ware, 2004; Illinsky & Steele, 2011):

- Our brains notice and process these instantly
- Color, form, movement, spatial positioning

Gestalt principles:

- proximity, similarity, continuity to signal relationships

Avoid clutter: too much visual noise reduces understanding.

Data Integrity (Tufte)

Decisions still need to be made

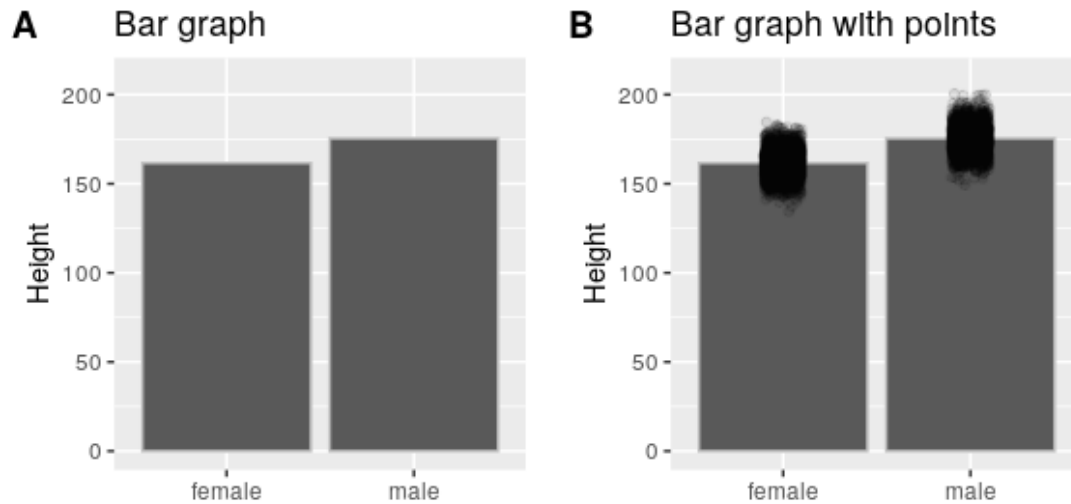


Figure 6.3: **Four** different ways of plotting the difference in height between men and women in the NHANES dataset. Panel A plots the means of the two groups, which gives no way to assess the relative overlap of the two distributions. Panel B shows the same bars, but also overlays the data points, jittering them so that we can see their overall distribution.

Decisions still need to be made

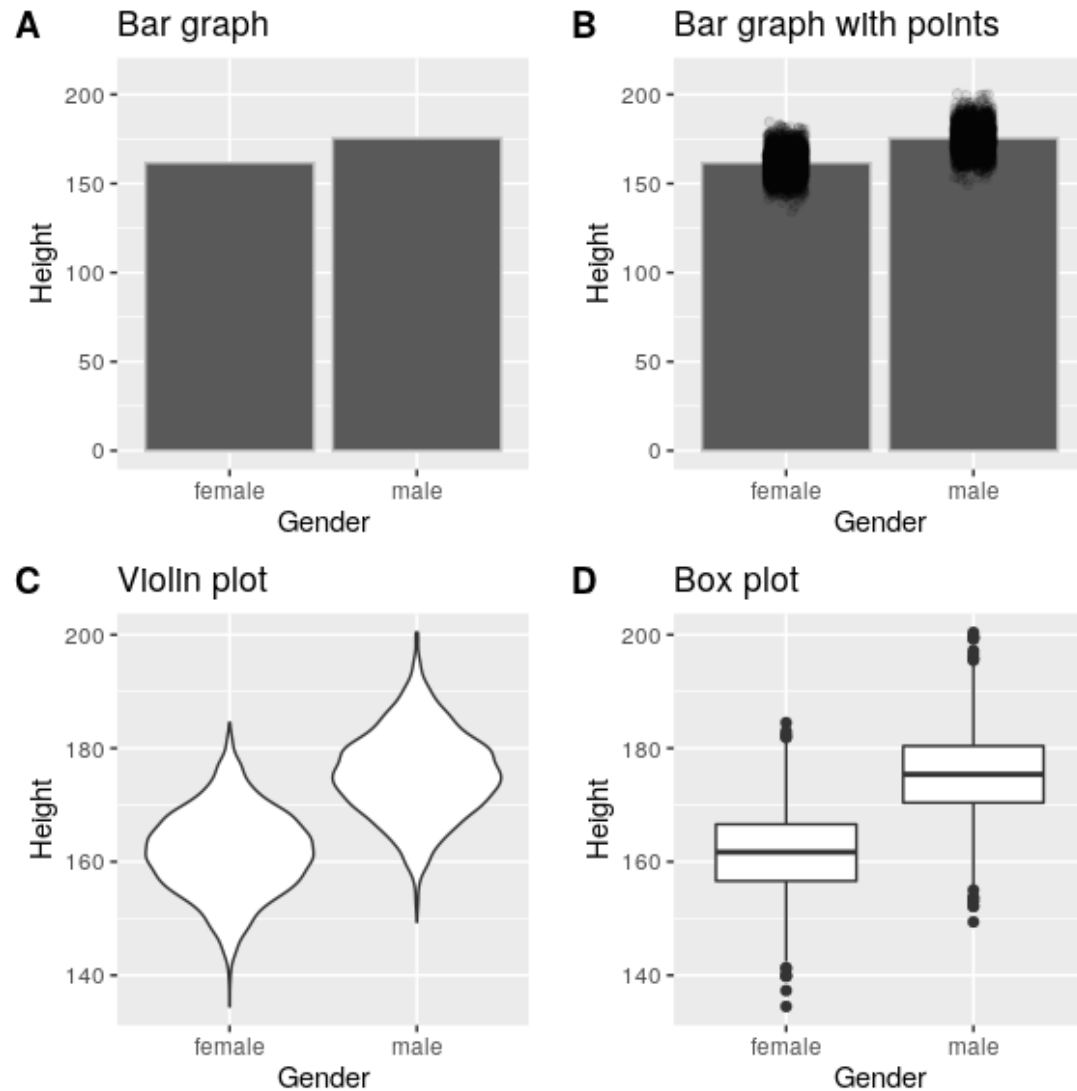


Figure 6.3: **Four** different ways of plotting the difference in height between men and women in the NHANES dataset. Panel A plots the means of the two groups, which gives no way to assess the relative overlap of the two distributions. Panel B shows the same bars, but also overlays the data points, jittering them so that we can see their overall distribution. Panel C shows a violin plot, which shows the distribution of the datasets for each group. Panel D shows a box plot, which highlights the spread of the distribution along with any outliers (which are shown as individual points).

Principles of good visualization*

Choose the Right Visualization for the Data	Clarity >>> novelty	• Use the correct graph type
Simplify	Remove Chartjunk & Maximize Data-Ink Ratio	• Keep it clean; Avoid unnecessary icons • Let the data shine; Use the least ink possible while ensuring clarity
Be Honest	Avoid distorting data & Misleading Scales	• Start axes at 0 • Maintain proportionality; don't exaggerate differences
Make Comparisons easy	Use multiple charts instead of overcrowding	• Maintain consistency across visualizations
Label Thoughtfully & Integrate Explanations	Place labels directly on data (instead of a separable legend)	• Combine text, graphics, and numbers for seamless storytelling
Show Context & Meaning	Highlight important trends while avoiding unnecessary clutter	• Include reference points, baselines, and annotations

- (Mostly) Edward Tufte
- “How to Lie with Statistics”- Darrell Huff
- “How Charts Lie”- Alberto Cairo
- “Calling Bullshit” – Carl T. Bergstrom & Jevin D. West

Data-Ink Ratio Checklist

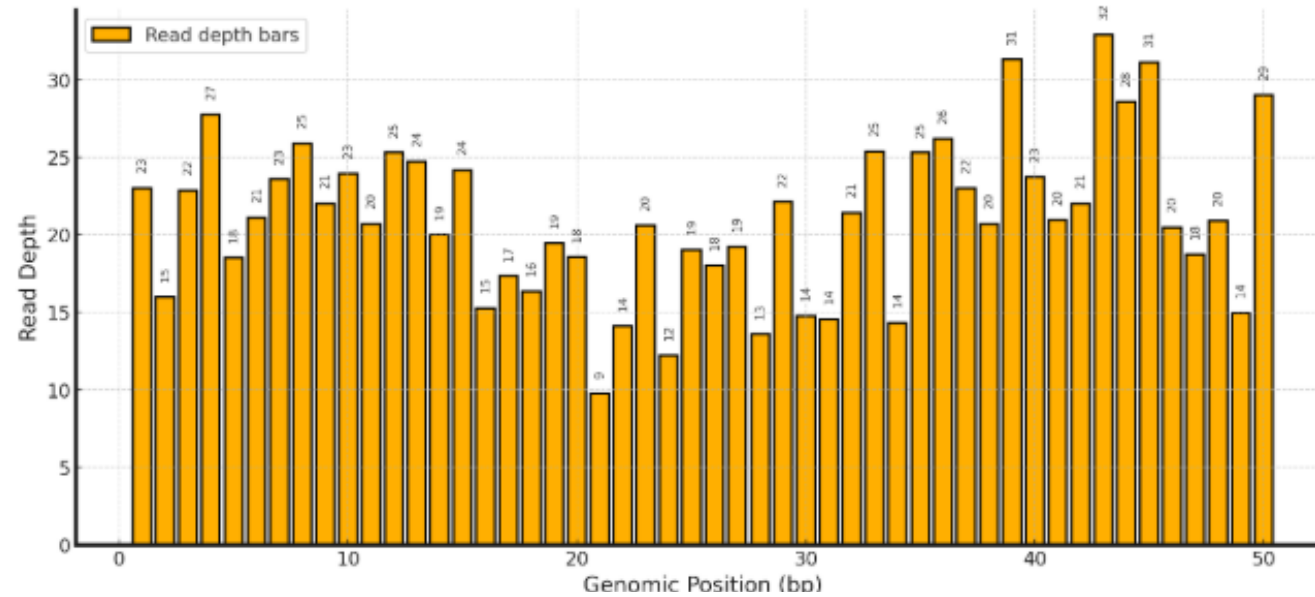
- SHOW THE DATA
- Minimize gridlines, borders, and background decorations
- **Avoid 3D effects**
- Directly label data when possible
- Reduce Cognitive Load

Color Is Not Decoration

- Use color intentionally: categorical vs. sequential
- Avoid red-green combinations for accessibility; use colorblind-safe palettes
- Don't use too many colors: **clarity > variety**

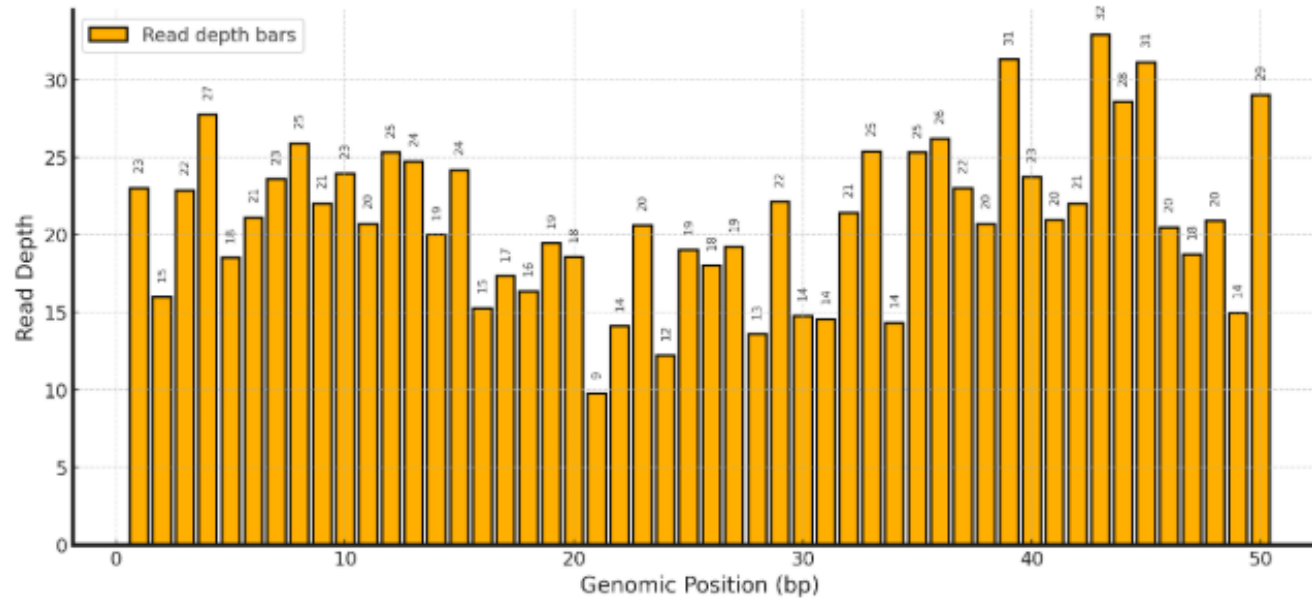
“Graphical excellence is what gives to the viewer the greatest number of ideas in the shortest time with the least ink in the smallest space.”

Genomic Coverage Across Locus XYZ (Bad Example)



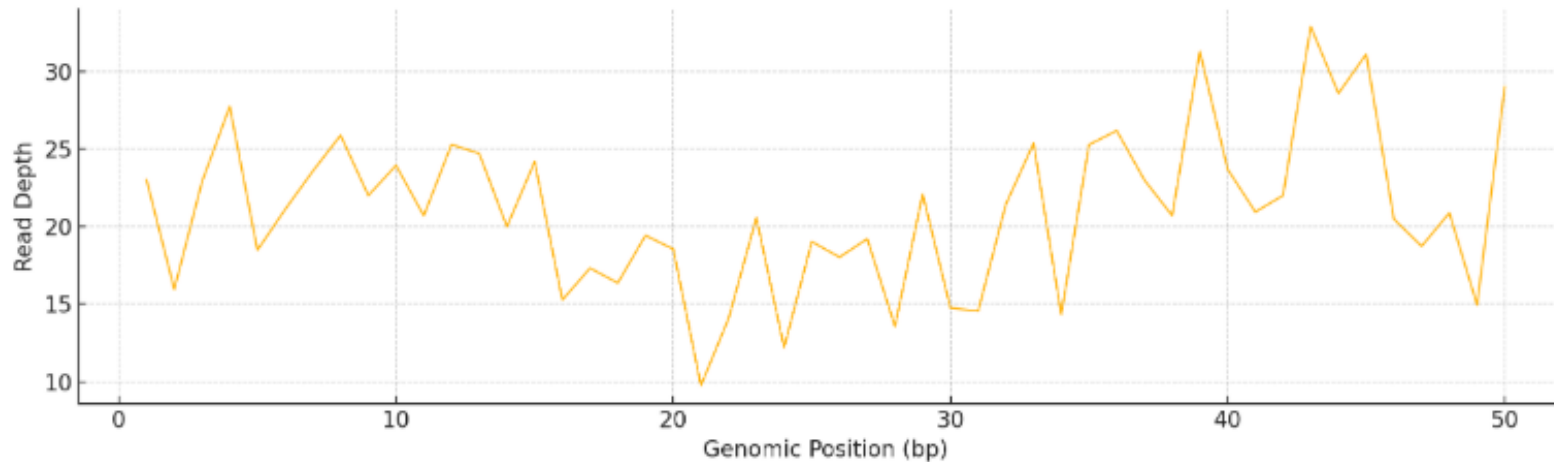
- **Low data-ink ratio:** most ink goes to decoration.
- **Chartjunk:** grid, 3-D-ish bars, bold title—all distract from the pattern.
- **Over-annotation:** numbers on every bar repeat the vertical axis.

Genomic Coverage Across Locus XYZ (Bad Example)



- **Low data-ink ratio:** most ink goes to decoration.
- **Chartjunk:** grid, 3-D-ish bars, bold title—all distract from the pattern.
- **Over-annotation:** numbers on every bar repeat the vertical axis.

Genomic Coverage Across Locus XYZ (Tufte-style minimalist)



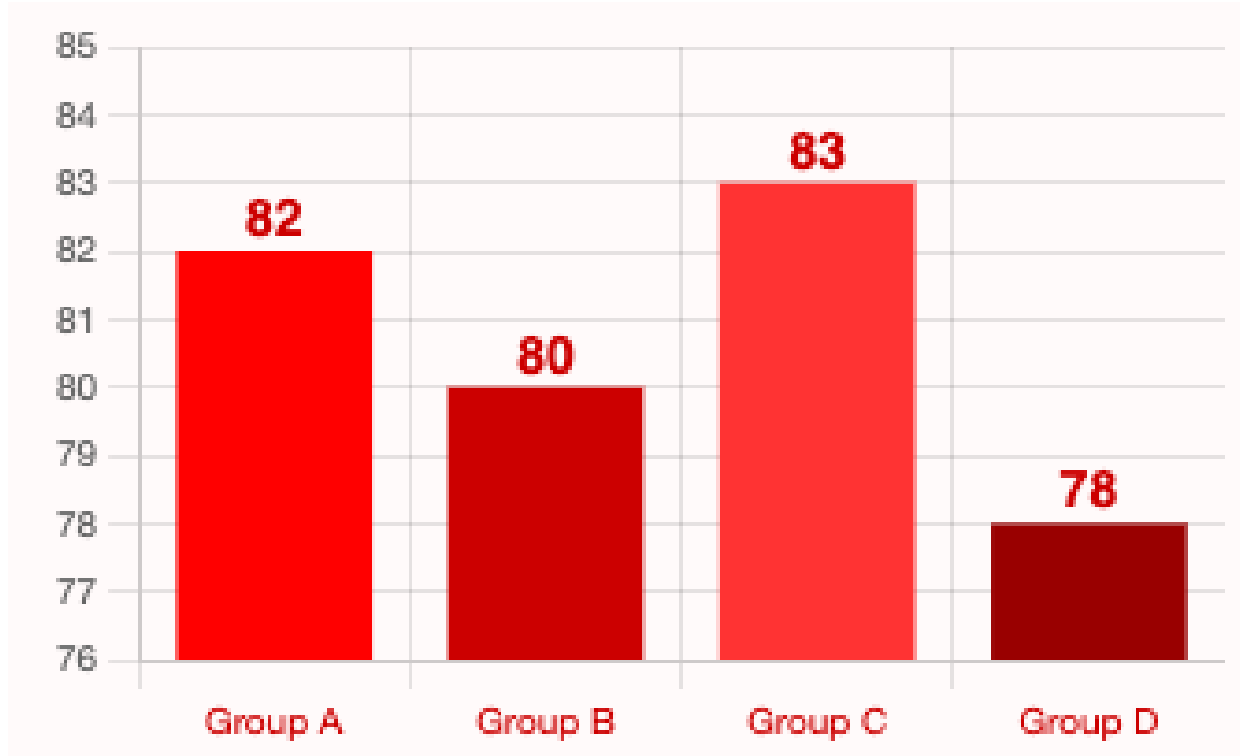
- **High data-ink ratio:** nearly all ink encodes data.
- **No chartjunk:** viewer sees the trend immediately.
- **Economy of annotation:** axis labels alone are enough; numbers can be read off the scale.

Graph Surgery: fix the visualization

For each graph:

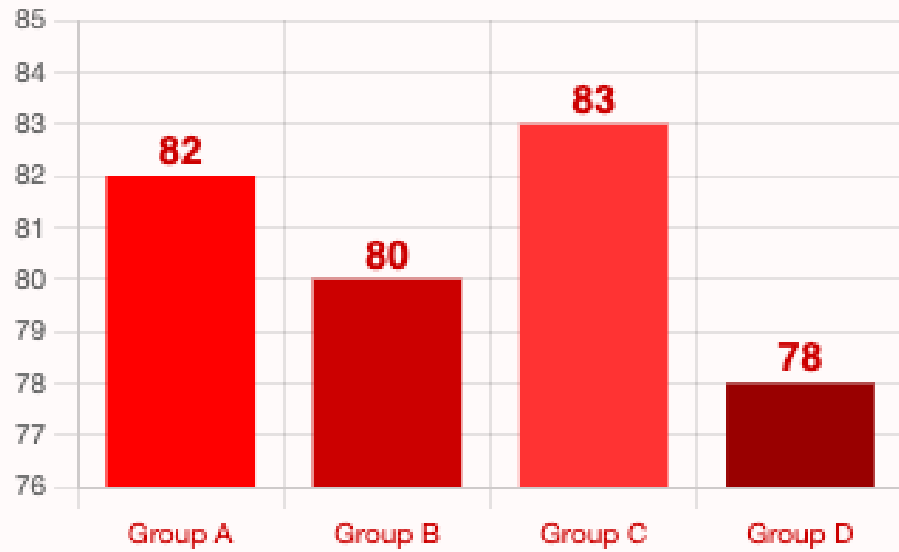
1. Identify the SINGLE biggest problem
2. Explain why it matters
3. Propose ONE repair

Example 1 - Diagnosis

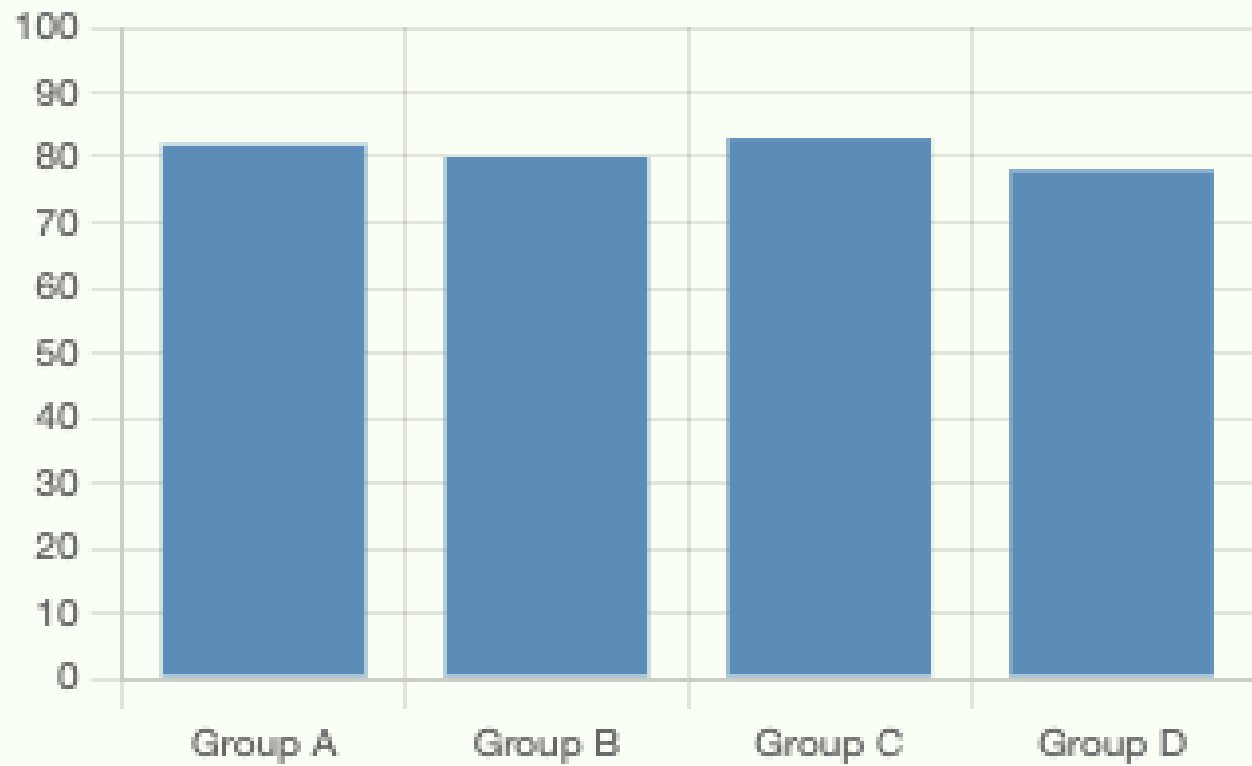


Example 1 - Remedies

Example 1 - Diagnosis



Example 1 - Remedies



Example 2 - Diagnosis



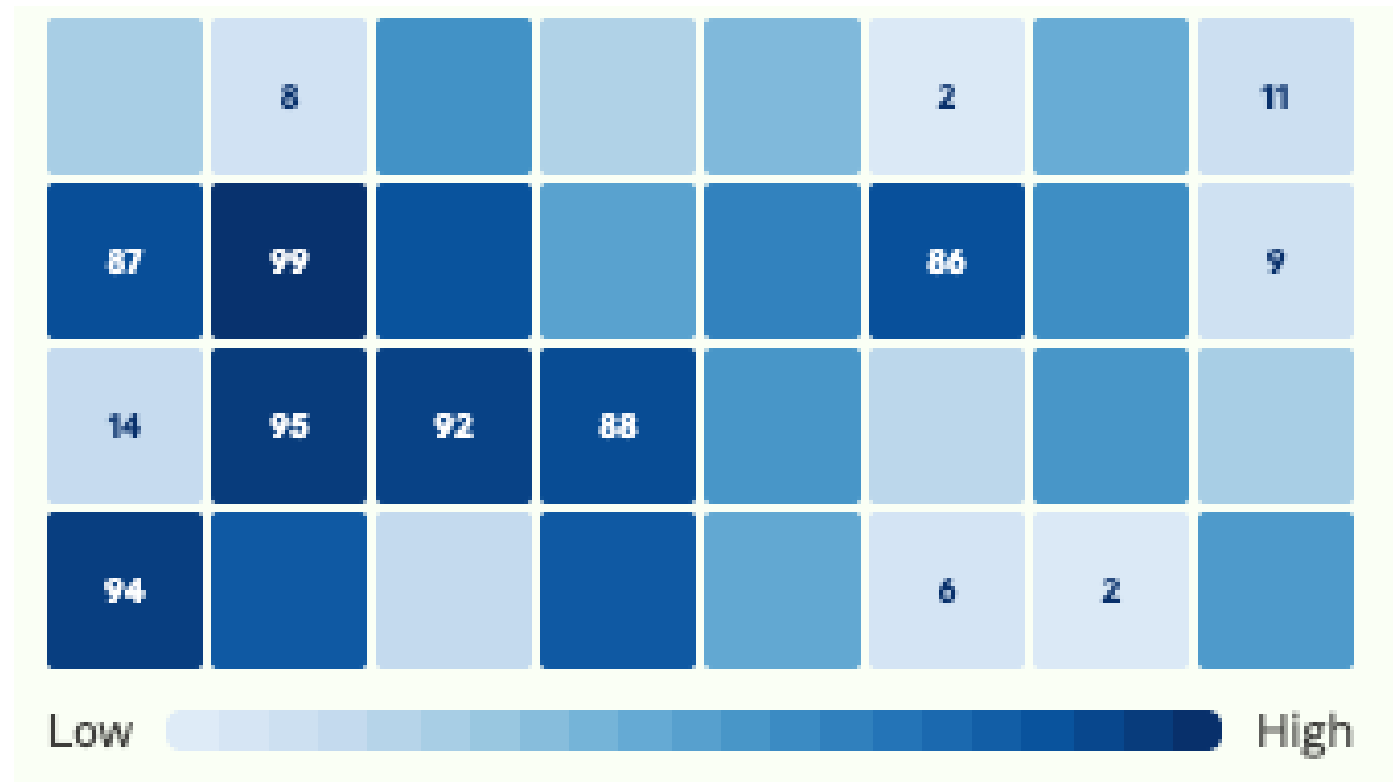
Example 2 - Remedies

Example 2 - Diagnosis

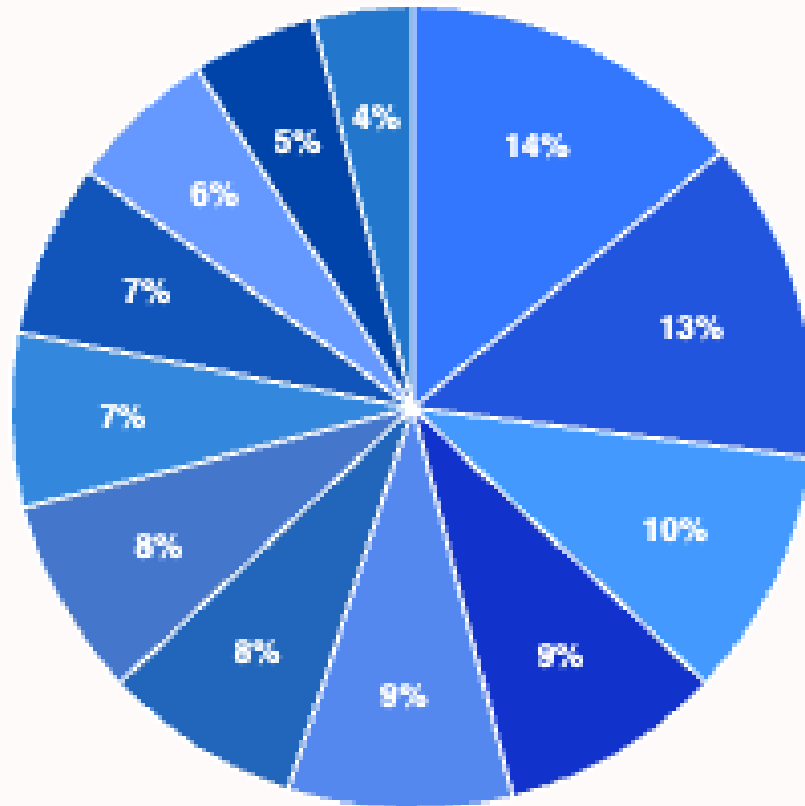


Example 2 - Remedies

Color should encode *meaning not decoration*

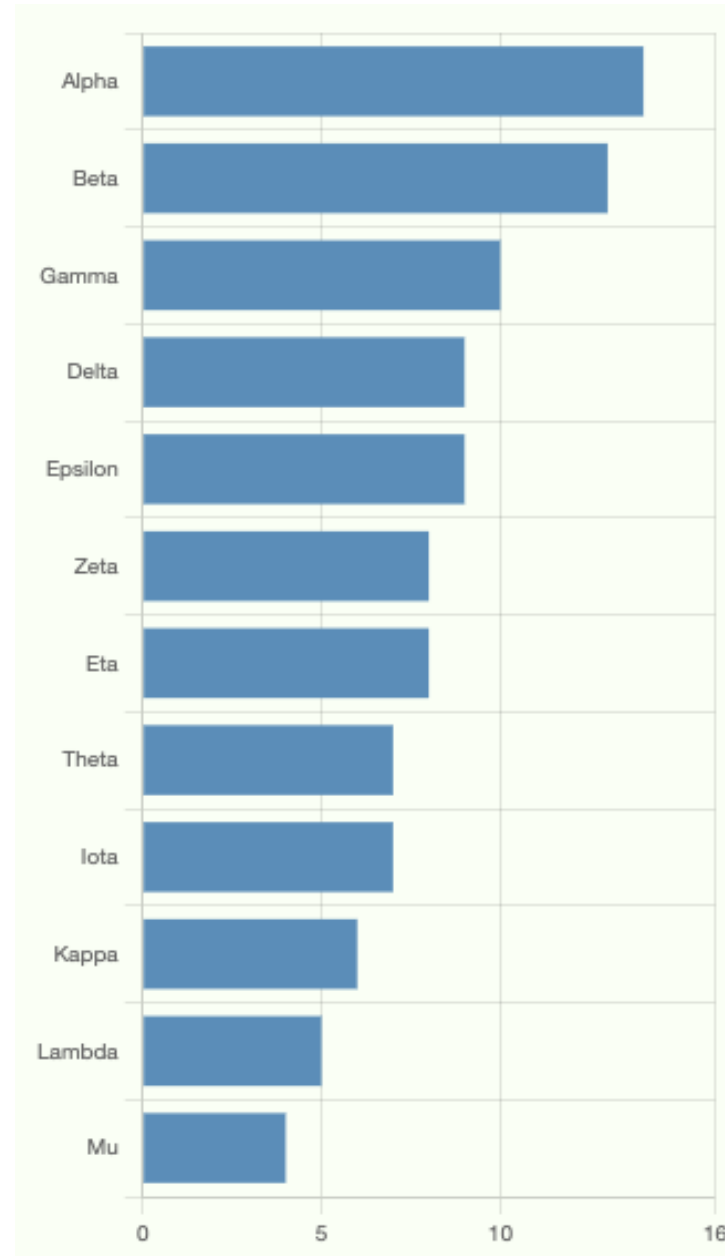
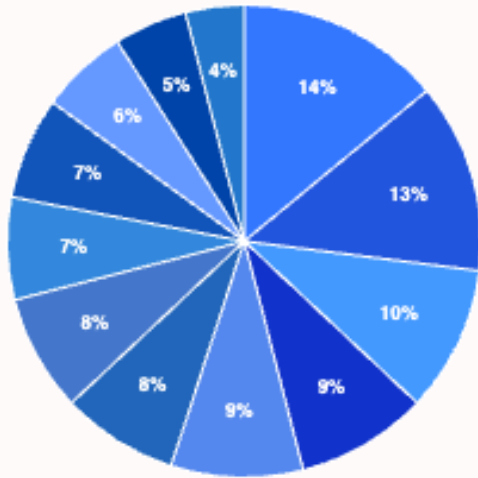


Example 3 - Diagnosis



Example 3 - Remedies

Example 3 - Diagnosis

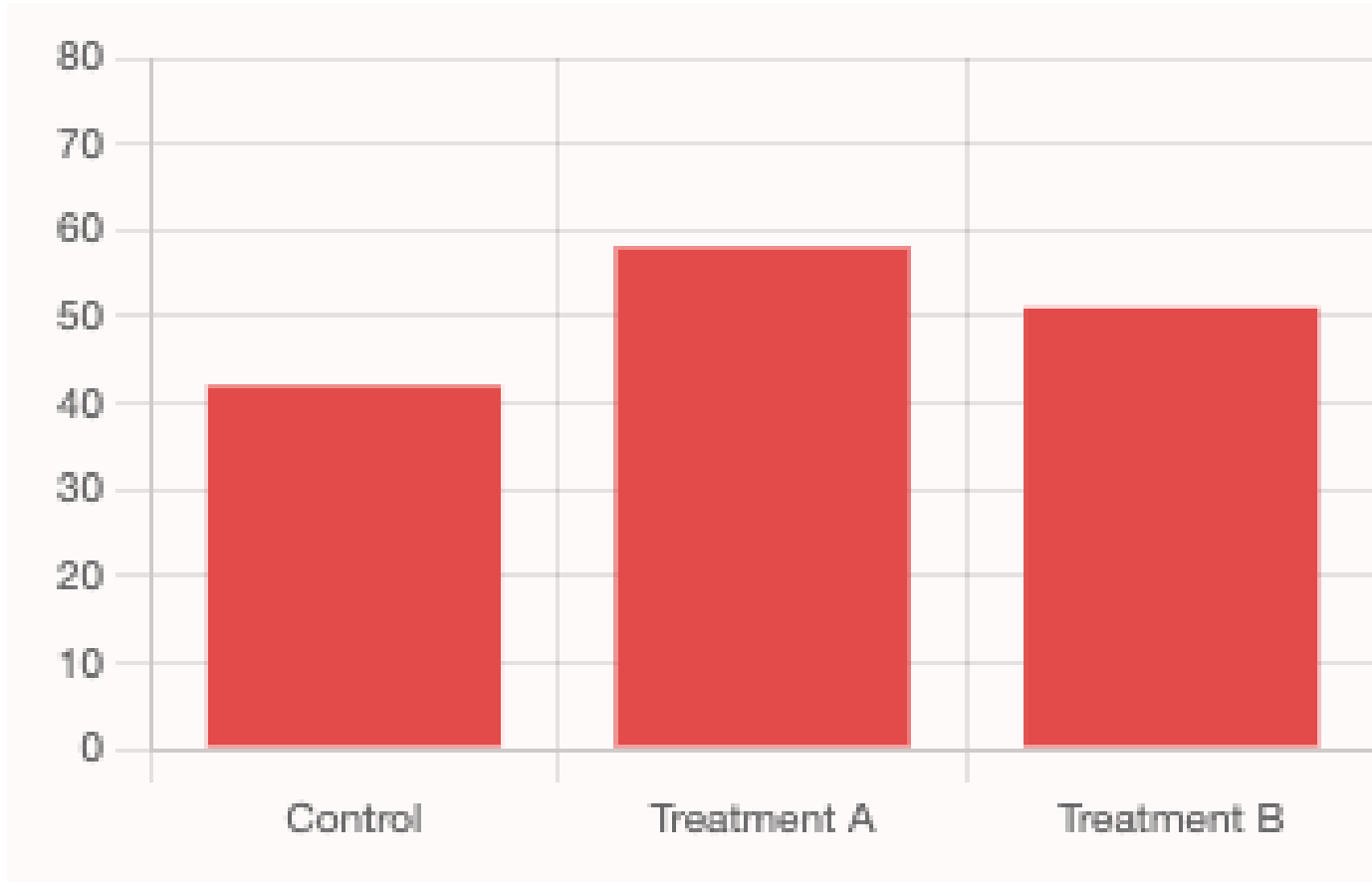


Example 3 - Remedies

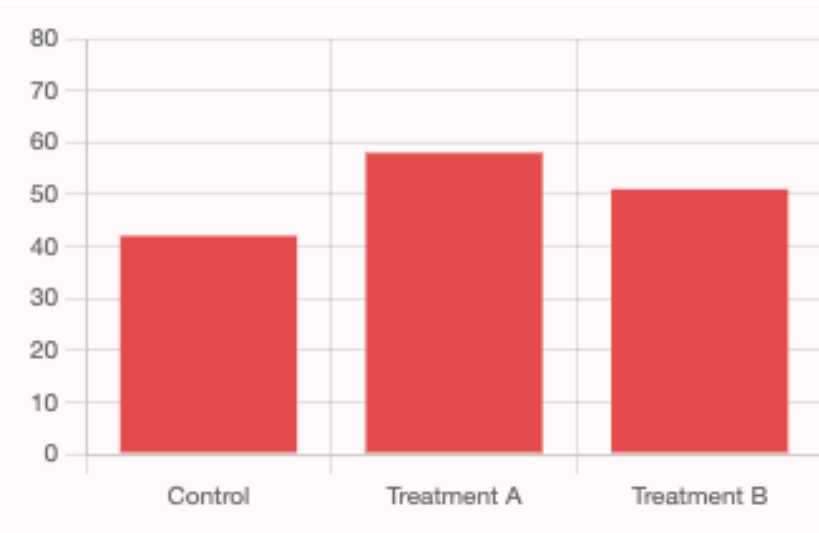
Biggest Problem

Example 4 - Diagnosis

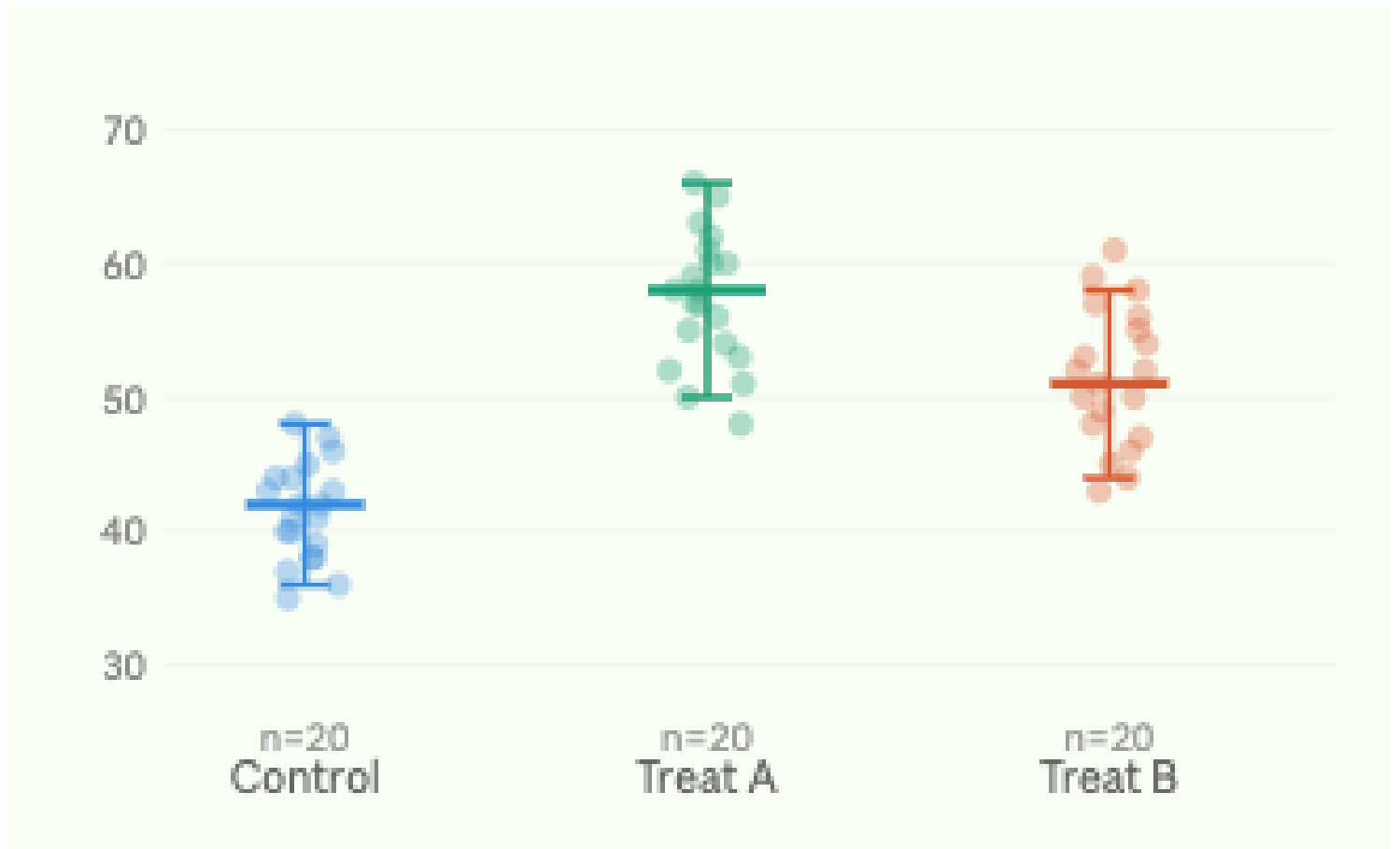
Example 4 - Remedies



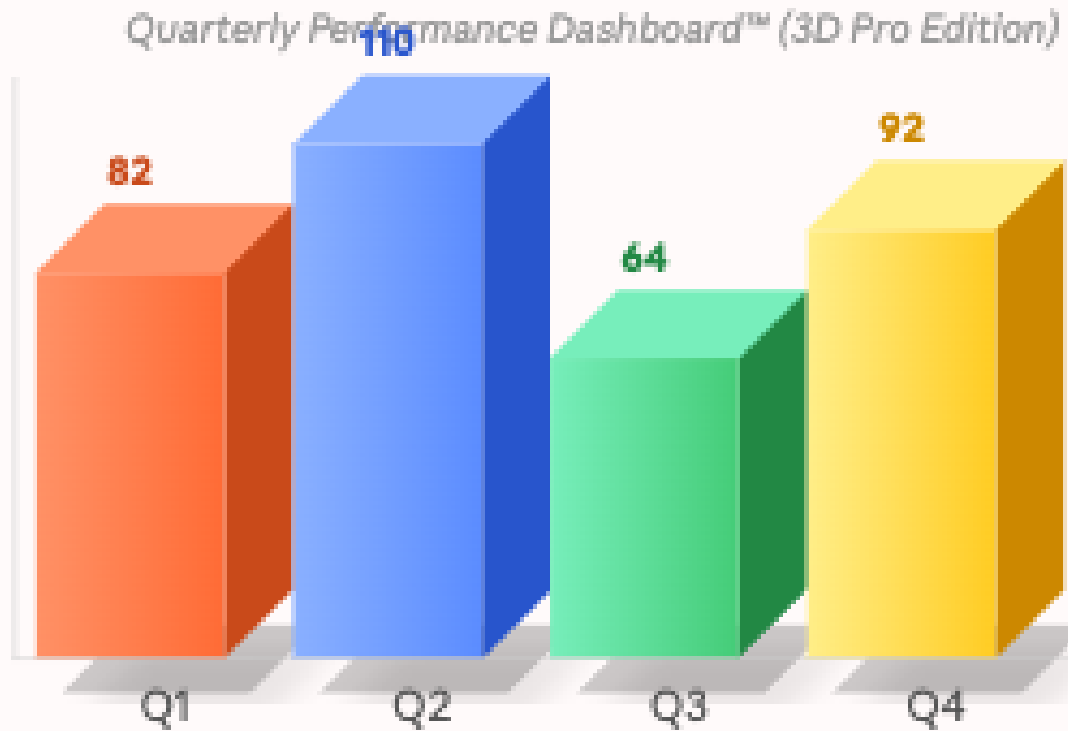
Example 4 - Diagnosis



Example 4 - Remedies

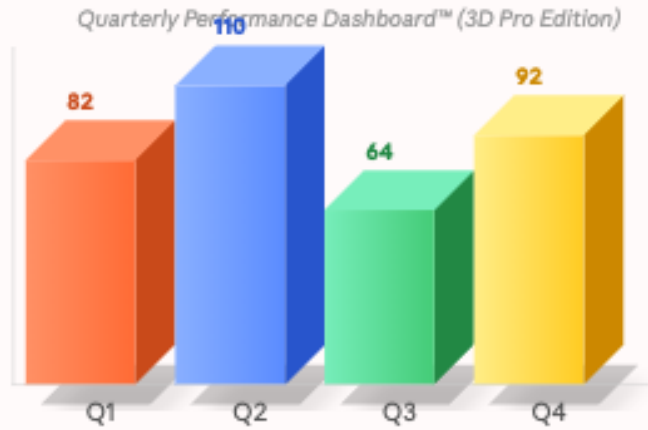


Example 5 - Diagnosis



Example 5 - Remedies

Example 5 - Diagnosis



Example 5 - Remedies



"Before submission" Checklist:

- ☐ What question does this figure answer?
 - ☐ Can someone interpret this figure in 5 seconds?
 - ☐ What is the main take-away?
 - ☐ Is variability visible? Is the axis honest?
 - ☐ Can we identify unnecessary information and simplify?
 - ☐ Is color adding information? Would a colorblind reader understand it?
 - ☐ Is the legend avoidable? Are fonts readable?
- 

Summary

1. Variable type & question-driven design leads to effective visuals
2. Cognitive principles guide audience attention and comprehension
 - Especially limit cognitive load
 - Emphasize relationships
3. Ethics and clarity are non-negotiable in data communication
4. Use visualizations as part of a larger narrative
 - If you don't give your audience a 'story', they will find one for you

Visualization is not decoration.

Visualization is scientific reasoning, communication, ethics, and persuasion combined.



References for Visualization Workshop:

1. Summary of 13 Charts: <https://people.rit.edu/wlrgsh/FINRobison.pdf>
2. Summary of Tufte's Challenger Argument: <https://williamwolff.org/wp-content/uploads/2013/01/tufte-challenger-1997.pdf>
3. Visual and Statistical Thinking: Cholera and The Challenger
https://public.websites.umich.edu/~prestos/Downloads/DC/Ritter_Tufte.pdf?ref=quirkos.com
4. R-package: <https://vizdator.wordpress.com/2015/05/06/space-shuttle-challenger-explosion-2/>
5. https://en.wikipedia.org/wiki/Flattening_the_curve
6. Claus Wilke Storytelling: <https://clauswilke.com/dataviz/telling-a-story.html>
7. Schimel "Writing Science": https://wp.optics.arizona.edu/kupinski/wp-content/uploads/sites/91/2023/01/Writing_Science_Joshua_Schimel.pdf
8. Hero's Journey Wikipedia: https://en.wikipedia.org/wiki/Hero%27s_journey
9. Three Act Story: <https://www.storyboardthat.com/articles/e/heroic-journey>
10. Statistics for the 21st Century:
[https://stats.libretexts.org/Bookshelves/Introductory_Statistics/Statistical_Thinking_for_the_21st_Century_\(Poldrack\)/06%3A_Data_Visualization/6.01%3A_Anatomy_of_a_Plot](https://stats.libretexts.org/Bookshelves/Introductory_Statistics/Statistical_Thinking_for_the_21st_Century_(Poldrack)/06%3A_Data_Visualization/6.01%3A_Anatomy_of_a_Plot)
11. Visual Display of Quantitative Information:
<https://kyl.neocities.org/books/%5BTEC%20TUF%5D%20the%20visual%20display%20of%20quantitative%20information.pdf>
12. Tidy Data: <https://r4ds.hadley.nz/data-tidy.html> and <https://vita.had.co.nz/papers/tidy-data.pdf>